DTU Physics

Department of Physics

Cogeneration of Power and Ammonia in Europe using Small Modular Pressurized Water Reactors

A Techno-Economic Analysis & Operational Strategy Optimization

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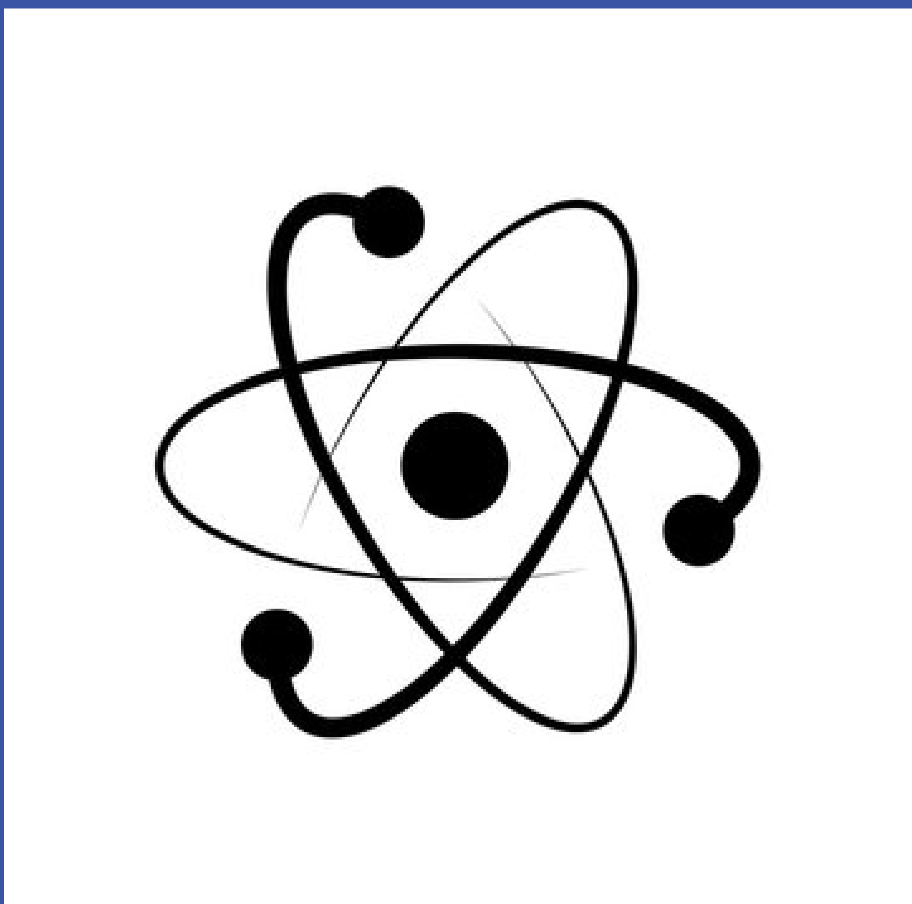
March, 2026



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Nomenclature

Abbreviations

ASU Air Separation Unit

BOAK Between First and N-th Of A Kind

BOP Balance Of Plant

CFR Cost and Freight

CS Carbon Steel

FCI Fixed Capital Investment

FOAK First Of A Kind

HB Haber Bosch

HTGR High-Temperature Gas Cooled Reactor

HTP High Pressure Turbine

HTSE High Temperature Steam Electrolysis

HX Heat Exchanger

IES Integrated Energy System

ITP Intermediate Pressure Turbine

LMTS Log Mean Temperature Difference

LTP Low Pressure Turbine

LTWE Low Temperature Water Electrolysis

LWR Light Water Reactor

MOC Material Of Construction

MOM Multi-unit Opex Multiplier

Ni Nickel

NOAK N-th Of A Kind

NPP Nuclear Power Plant

O&M Operations and Maintenance

PSA Pressure Swing Adsorption

PWR Pressurized Water Reactor

SG Steam Generator

SMR Small Modular Reactor

SOEC Solid-Oxide Electrolysis Cell

SS Stainless Steel

TLR Technology Readiness Level

TPB Triple-Phase Boundary

TPD Tons Per Day

VHGR Very High-Temperature Gas Cooled Reactor

Physical Constants

F Faradays constant $96\,485.332\,\mathrm{s\,A\,mol^{-1}}$

R Universal gas constant $8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$

Parameters

ΔG Gibbs free energy $\mathrm{kJ\,mol^{-1}}$

ΔH Enthalpy change of reaction $\mathrm{kJ\,mol^{-1}}$

ΔS Entropy change of reaction $\mathrm{J\,mol^{-1}\,K^{-1}}$

ΔT_{pinch} Pinch point temperature difference

$\dot{Q}_{SG}$ Steam Generator Thermal Power kW

$\dot{W}_{net}$ Net system power output kW

$\dot{W}_{stack}$ SOEC stack power consumption kW

π Profits €/y

C_{BM} Bare Module Cost €

C_{OL} Cost Of Labor €

F_p Pressure cost factor

j Current density $\mathrm{A\,cm^{-2}}$

K_α	Equilibrium constant	
$r_{\mathrm{NH_3}}$	Ammonia reaction rate	
V_{Nernst}	Nernst Potential	V
V_{op}	Operating Voltage	V
V_{th}	Thermoneutral Voltage	V

Approval

This thesis has been prepared over six months at the Department of Physics, at the Technical University of Denmark, DTU, in partial fulfillment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of thermodynamics, energy systems, energy economics, and nuclear power generation.

Henrik Stenrud Bergersen - s233194

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Signature

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Date

Abstract

Most of today's global ammonia synthesis relies on fossil fuels, accounting for around 2% of global CO_2 emissions, based on the Steam Methane Reforming technology. While production largely serves the fertilizer industry, demand is projected to grow as ammonia expands into the low-carbon fuel market. Nuclear power offers a viable pathway to decarbonize this energy-intensive sector while diversifying revenue streams. Specifically, emerging small modular reactors (SMRs) are attracting attention for their site flexibility, making them ideal partners for integrated ammonia plants. To increase profitability, an SMR-driven cogeneration power and ammonia system could potentially allocate the produced thermal energy for ammonia production during low electricity prices, making returns less sensitive to market fluctuations.

This study investigates the techno-economic potential of deploying small modular pressurized water reactors for combined power and green ammonia production within the European power market. Utilizing a plant-scale thermodynamic model and revenue projections, the results indicate that by bridging the ammonia and electricity markets, the facility can avoid selling electricity at a loss during price troughs while capitalizing on peak demands. Furthermore, the analysis demonstrates that for a nominal capacity of 500 tons of ammonia per day and an ammonia spot price of 700 €/t, ammonia production is more profitable than power production over 90% of the operating hours in the Norwegian market, 80% in the Swedish, and 70% in the Finnish market.

Acknowledgements

For the completion of this thesis, I would like to thank my internal supervisors for their invaluable guidance, technical expertise, and constructive feedback throughout the entire research process. I would also like to thank my external supervisor at AFRY for the interesting insights into the development of nuclear projects and industrial practices.

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1 Introduction

1.1 Background

In recent years, interest in nuclear power has significantly increased globally, driven by a growing power demand and desire for climate change mitigation, leading to increased investments and policy support for nuclear energy expansion [1]. Furthermore, the Small Modular Reactors under development have attracted considerable attention due to their potential for factory-based modular construction, which could lower both production and on-site assembly costs, as well as increase site flexibility [2].

SMRs are generally defined as nuclear power reactors with electrical outputs between 10 and 300 MWe, covering a broad range of technologies within this capacity window. They are characterized by several technical features that aim to improve construction predictability and reduce both project costs and delivery times compared to conventional large nuclear reactors. Their smaller size means they cannot exploit economies of scale and instead, their competitiveness relies on serial production. Consequently, SMR development is expected to benefit from steeper learning curves driven by greater modularization, design simplification, and standardization.

As nuclear reactors can deliver both power and process heat directly, SMRs could benefit from expanding into additional energy markets by integrating them into cogeneration systems. Cogeneration has the potential to increase the overall energy utilization by providing both electricity and heat, enabling participation in two different commodity markets and potentially enhancing plant profitability as well [3]. Due to their high ratio of fixed-to-variable costs, nuclear power plants (NPPs) are traditionally utilized as baseload assets. Currently, load-following operations, whereby NPPs reduce output to match fluctuating demand, result in the underutilization of carbon-free energy and can induce detrimental thermo-mechanical stress on reactor components. Unlike gas-fired turbines, which achieve fuel savings during periods of lower output, NPPs realize negligible operational savings at reduced power due to their fixed cost structure. In addition to capital expenditures, Operation and Maintenance (O&M) costs remain largely independent of the power level. By reallocating energy toward applications with higher economic value, such systems can potentially sustain high operational output, while regulating the power delivered to the grid [4].

The feasibility of such an Integrated Energy System (IES) with SMRs as the power source depends on the maturity of the secondary commodity market, as this must provide a competitive revenue stream for the project. Ammonia production represents one of the most energy-intensive industrial processes in Europe, and decarbonizing these industries is a central challenge for Europe's net-zero ambitions. Therefore, the ammonia market is a promising option for a cogeneration system, where nuclear power can offer high-capacity carbon-free energy. Ammonia synthesis is a highly electricity and heat-intensive process, with typical energy demands in the range of 30-50 GJ per tonne of ammonia produced [5]. It is a strongly reducing compound and ranks among the most widely produced chemicals globally, with about 70-80% of its output consumed by the fertilizer industry [6]. The role of ammonia is also expected to expand beyond fertilizers, with growing demand linked to its use as a low-carbon energy carrier, energy storage medium, and fuel for transportation and power generation.

Currently, most of the global ammonia synthesis is based on fossil energy, and the economics of ammonia are strongly linked to natural-gas markets. The dominant production method is the Haber–Bosch (HB) process, and hydrogen feedstock is traditionally derived from steam methane reforming, which accounts for most of the natural-gas consumption in the process[6]. Recent industry projects and policies, including green ammonia initiatives and pilot electrolyzer plants, demonstrate growing commercial interest in establishing low-carbon ammonia value chains in Europe. Evaluating the techno-economic trade-offs is therefore timely for assessing whether SMR-based cogeneration can be competitive and supportive of Europe's industrial decarbonization of ammonia production.

Assessing the revenue potential of this system within the European energy market requires integrating simulated production rates with regional market prices to define an optimal operational strategy for maximizing annual returns. To evaluate the system's economic viability, these results are benchmarked against a standalone nuclear power-only scenario. The findings may be valuable to both future research on nuclear-ammonia system integration and decision-making for companies or investors considering investment in projects involving nuclear-powered cogeneration and/or ammonia production.

1.2 Objective and research question

This research aims to investigate the techno-economic potential of deploying small modular pressurized water reactors (PWR-SMR) for cogeneration of power and green ammonia within the European power market. The study focuses on increasing the plant's profitability by adjusting the electrical output in response to market conditions while integrating heat and electricity for ammonia production. By developing a steady-state model of a small to medium-scale nuclear cogeneration facility, the production capacities and energy requirements are first established. Next, a cost estimation framework is used to determine the system's individual production costs of electricity and ammonia, functioning as the basis for the profitability analysis. By leveraging the power plant's energy output for ammonia production during periods of low or negative electricity prices, the study aims to examine the overall economic performance of the system and the business case for cogeneration-based investments. A key aspect of the analysis is the dimensioning of the system and determining the optimal allocation of thermal and electrical output between the power grid and the nearby ammonia production facility. The thesis will explore the mode of operating conditions under which cogeneration becomes economically viable, considering system efficiency, thermal utilization, and power market volatility. Lastly, scaling effects in investment and production are discussed.

Research questions

To guide the study towards concrete results, two central research questions have been formulated. The technical viability of a nuclear-integrated ammonia facility depends on the coupling of the reactor's thermal output with the chemical plant's requirements. The physical dimensioning of components and the synthesis loop must be considered. Therefore, it is necessary to evaluate the configurations and heat-integration strategies to match an SMR's specific output. This leads to the first research question. To optimize the plant's revenue, the system must balance the sales of electricity and ammonia. This decision depends on the fluctuating price ratio between the two commodities. By analyzing the opportunity cost of electricity relative to the market value of ammonia, this study captures how market volatility dictates the most profitable operational mode. This leads to the second research question:

1. What is the required system dimensioning to achieve a self-sufficient energy balance for both thermal and electrical demands of the cogeneration system, and how does cogeneration influence system efficiencies?
2. What is the marginal break-even revenue point for the nuclear cogeneration facility, and how do regional variations across European energy markets impact the system's operational strategy

1.3 Review of analyses on relevant nuclear power cogeneration

Cogeneration has been widely studied in the literature as a strategy to enhance the profitability of nuclear power, both through retrofits of existing plants and in the design of new reactors. This focus has emerged in light of the economic challenges confronting today's nuclear fleet and the growing momentum behind advanced reactor technologies. Numerous theoretical studies on nuclear-hydrogen cogeneration have been conducted, involving the integration of both conventional light-water reactors (LWR) and new-generation SMRs. The following studies have been identified as relevant to the scope of this research.

[7] investigated the techno-economic potential of cogenerating electricity and hydrogen from a NPP using high-temperature steam electrolysis (HTSE). The assessment covers both the technical configuration and cost estimations for hydrogen production, where the HTSE process is based on solid-oxide electrolysis cells (SOECs), consuming steam and electricity from a pressurized water reactor (PWR). It focuses on retrofitting an existing PWR to integrate hydrogen production and analyzes the integration by developing a model in the ASPEN/HYSYS software. The results indicate that significant gains are achievable through market diversification. By diverting energy between power and hydrogen production, the nuclear facility can effectively hedge against periods of low electricity prices while remaining positioned to capitalize on high-price volatility

[8] studies the economic feasibility of various hydrogen production and storage systems coupled with a High-Temperature Gas-cooled Reactor (HTGR) based on the levelized cost of hydrogen (LCOH). The evaluated production methods include SOEC, thermal decomposition of methane, and decomposition of ammonia, while the storage methods considered are compressed hydrogen, liquid hydrogen, and ammonia conversion. Among these systems, the authors identify the configuration combining SOEC for hydrogen production and compression for storage as the

most economical, assuming an HTGR lifetime of 10 years. To benchmark performance, all production systems were compared to a PWR coupled with low-temperature water electrolysis (LTWE), using typical nuclear lifetimes as a reference. The results indicate that the LCOH from a PWR–LTWE system with a 40-year lifetime is lower than that of an HGTR–SOEC system if the HGTR lifetime is below 15 years. For a PWR-LTWE system with a lifetime of 60 years, the HGTR needs to have a lifetime of 35 years to be more economical.

[9] assessed the techno-economic potential of SMR deployment for non-electric applications, such as synthetic fuel production and industrial processes in the U.S. The dissertation concludes that while SMRs are generally not competitive for electricity generation alone, they can become profitable when applied to industrial processes and show strong potential for non-electric applications such as hydrogen and ammonia production. In particular, it finds that SMR deployment for hydrogen production under the U.S. Hydrogen Production Tax Credit (H2 PTC) often results in negative break-even prices, meaning the subsidies exceed the cost of production.

[10] proposed a novel ammonia-hydrogen-electricity polygeneration system, using a very high-temperature gas-cooled reactor (VHTR) with an Iodine-Sulfur (IS) cycle for thermochemical hydrogen production. The study highlights that employing ammonia as a hydrogen carrier helps overcome several challenges related to hydrogen storage, transportation, and safety. The reactor coolant is helium, which provides high-temperature heat for hydrogen production, low-temperature heat for power generation, while the high-temperature oxygen coming from the IS cycle, is used in the HB process. Results demonstrate that the high-grade heat supplied by the reactor enables comparatively high conversion efficiencies. Furthermore, integrating ammonia synthesis into the hydrogen–electricity cogeneration system increased the overall energy efficiency by 7.5% relative to the case without ammonia production.

In 2024, Sadeghi et al. [11] did a comprehensive study of the thermal integration and economics for nuclear-hydrogen cogeneration via high-temperature steam electrolysis driven by a light-water reactor based on the VVER-1000 NPP. They focused on minimizing the power loss factor by investigating the optimal point of steam extraction from the NPP, which can also minimize the HTSE costs. For safety reasons, the proposed system also incorporates an intermediate loop between the NPP and the HTSE with a proper pressure gradient, acting as a radioactive barrier. The authors simulated several possible scenarios for steam extraction and found that steam extraction decreases the electrical efficiency of the NPP, but increases the total cogeneration efficiency by up to 5% for a large-scale HTSE plant. Furthermore, they showed that the operation and maintenance cost of the system has the largest share among all cost components due to the electricity consumption rate of SOEC stacks.

While much of the existing literature on nuclear cogeneration focuses on hydrogen production alone, this thesis extends the system boundary to include ammonia production, as in the first paper presented above. Since ammonia synthesis relies on hydrogen as a feedstock, the analysis captures both the hydrogen production step and its direct industrial application. Regarding the choice of reactor technology, the advanced SMR concepts employing novel moderators and coolants may ultimately achieve higher power outputs or deliver higher outlet temperatures. However, their commercial deployment is likely to be behind that of LWR-based SMRs. For example, the demonstrated average operating lifetime of commercial-scale PWRs is 40-60 years, while for HGTRs it is approximately 9 years [12][13]. Moreover, GE Hitachi concluded that the key factor influencing HTSE performance when coupled with nuclear power is not the reactor outlet temperature, but rather the levelized cost of electricity of the reactor design [14]. Therefore, building on previous studies, there is potential to investigate a cogeneration system based on evolutionary SMRs that draw experience on proven technologies.

1.4 Company description

This master's thesis project has been carried out under the guidance of AFRY Energy – Nuclear Norway, the Norwegian branch of AFRY's Nuclear division. The unit specializes in nuclear energy technology services and operations in the Norwegian context.

AFRY is a Swedish–Finnish engineering, design, and advisory company with a global presence. AFRY Nuclear Norway collaborates with both private corporations and governmental agencies, providing consultancy in areas such as management, safety and security, radioactive waste handling, construction, technical documentation, and the decommissioning of nuclear facilities[15].

1.5 Limitations and Scope

This thesis is not based on a site-specific case study but is instead framed within the context of the European energy market, as the relevant market prices are derived from this continent. While certain locations may be more suitable than others for implementing such a system due to factors such as energy demand, geographical characteristics, or industrial integration, detailed site-specific analyses are beyond the scope of this work.

Integrating nuclear thermal energy with ammonia synthesis necessitates a rigorous approach to process safety. The inclusion of electrochemical and chemical synthesis introduces hazard profiles distinct from conventional nuclear power generation. Specifically, the storage and transport of hydrogen present risks of leakage and potential deflagration [10]. Notably, the downstream conversion of hydrogen into anhydrous ammonia can serve as a strategic method to mitigate some of these gaseous storage risks. While these safety considerations are critical to system deployment, the detailed design of safety instrumented systems remains beyond the scope of this investigation.

To maintain a manageable computational load and model complexity, the simulation is limited to the core systems spanning from the steam supply interface to the final liquid ammonia product. Consequently, the nuclear reactor is represented via a simplified secondary steam cycle, consisting of steam turbines, a condenser, feed pumps, and a heat exchanger acting as the steam generator. Furthermore, both intermediate and final storage of the synthesized chemicals are not part of the analysis.

To keep the scope focused, the techno-economic evaluation is limited to analyzing revenue streams and production costs from the perspective of the energy producer

It is also important to note that new information on topics regarding new reactor technologies continues to evolve throughout the course of this study. Therefore, the latest data on economics, operations, and regulations may not be up to date at this moment.

2 Theory

This chapter establishes the theoretical framework for the proposed cogeneration system and the subsequent techno-economic analysis. It delineates both the technical foundations and the primary economic drivers governing the system's performance. The discussion commences with an overview of SMR technology and the specific reactor design employed in this study, followed by a detailed review of green ammonia synthesis via HTSE. Subsequently, the configuration of the integrated system is examined with a breakdown of the operational expenditures and revenue streams that define the economic model.

2.1 Small Modular Pressurized Water Reactors

The concept of small modular reactors has been adopted internationally, yet the designs differ substantially in coolant choice, fuel type, technology readiness level (TRL), and licensing readiness level (LRL). Deployment approaches also vary, ranging from single-module installations to multi-module plants, as well as mobile configurations such as floating barge-mounted units. The extent of modularization likewise differs across designs.

At present, roughly 11 PWR-SMR designs and concepts exist worldwide, most still in development, though several are planned for near-term deployment. While alternative concepts such as molten salt or gas-cooled SMRs are also under consideration, water-cooled modular designs remain the most prominent. These reflect both the early design choices supported by fuel and component suppliers and the extensive body of operating experience gained over more than 60 years, the vast majority of which has been with water-cooled reactors[16]. Although generally viewed as a new, developing technology for commercial grid power supply, it is important to acknowledge that the concepts of PWR-SMR or integral pressurized water reactors (iPWR) date back to nearly the beginning of commercial nuclear energy. The shipping industry and navy, especially, have been a driver for the innovation of compact reactors since the 1950s[17].

In recent years, SMR development has increasingly focused on iPWR designs, which incorporate the main primary system components within a single containment. The figure below illustrates the generic design of a typical two-loop PWR and an iPWR.

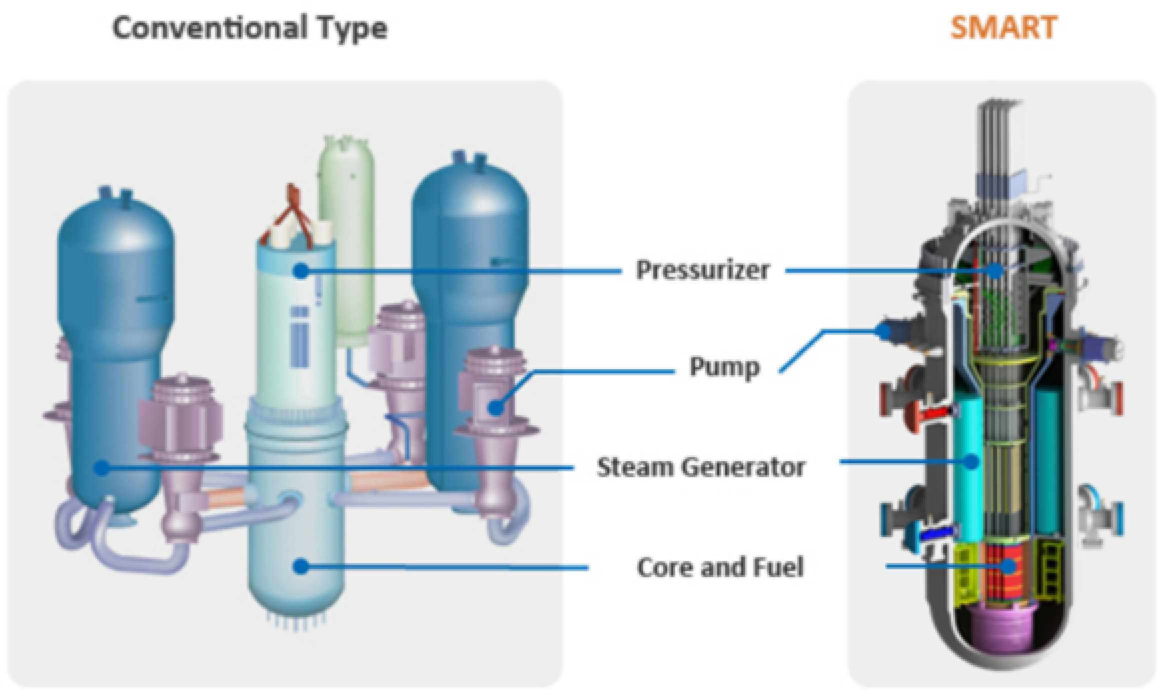


Figure 2.1: Conceptual diagram comparing a conventional full-size PWR with the South Korean SMART PWR design [18]

In the reactor core of a PWR, fuel assemblies containing slightly enriched (3.3 %) UO_2 undergo fission, releasing high-velocity neutrons and converting a fraction of their mass into energy. To increase the fission probability, these fast neutrons are moderated by typically pressurized light water, which reduces the neutrons' kinetic energy to the thermal energy range (0.025 - 1eV). This water serves a dual purpose as both a moderator and a coolant. To regulate the neutron population and reactivity, the system employs several control mechanisms, including boric acid concentration adjustment and the insertion of control rods, both being strong neutron absorbers. Unlike a Boiling Water Reactor (BWR), in a PWR, a pressurizer maintains the primary coolant under about 150 atmospheres to prevent bulk boiling at the typical core outlet temperature of 300-340° C. Within the steam generator, thermal energy is transferred to a secondary loop, where separate feed water is evaporated to steam. This steam expands through a series of turbines to drive a generator, converting the thermal energy into electric power. Finally, the exhausted low-pressure steam is liquified in a condenser, closing the Rankine cycle [19]. The main difference between a conventional PWR and the smaller iPWR is that many of the components in the secondary system are situated together with the reactor inside the same pressure vessel, as illustrated in figure2.1. Because the core is also smaller, some iPWR designs require slightly more enriched uranium, but below the high assay low enrichment uranium (HALEU) threshold at 5% [1]. Furthermore, these next-generation PWR-type SMRs often incorporate enhanced safety architectures designed to mitigate component failure and human error. A primary feature is the integration of passive safety systems, which rely on natural physical phenomena.

2.2 Nuclear powered ammonia production

Ammonia synthesis

The most widely used pathway for fossil-free ammonia synthesis builds upon the Haber–Bosch process, combined with water electrolysis, replacing the conventional steam methane reforming. The process synthesizes ammonia by reacting nitrogen with hydrogen under elevated temperatures (350–550 °C) and pressures (100-450 bar) [20], typically in the presence of an iron-based catalyst. The reaction, shown below, is exothermic and equilibrium-limited, requiring optimized conditions to balance conversion efficiency and economic feasibility[6].

$$3\,H_2 + N_2 \rightleftharpoons {}_2NH_3 \qquad \Delta H_{rx} = -92.44\,\mathrm{kJ\,mol^{-1}}\,N_2$$

In a typical industrial ammonia synthesis process, the nitrogen and hydrogen feed streams are first compressed and mixed with the recycled gas. This combined stream is then introduced into a multi-bed reactor, where around 15–20 mol% of the feed is converted to ammonia. The reactor effluent is subsequently cooled, and ammonia is separated from unreacted nitrogen and hydrogen by condensation at -20 to 30 °C. Due to the significant vapor pressure of ammonia at these conditions, about 2–5 mol% of ammonia remains in the gas phase and is returned to the reactor with the recycle stream [20]. A purge stream is typically required to prevent the buildup of inert gases present in the synthesis loop [21]. A general diagram of the HB-process is presented in figure 2.2.

The catalyst selection has only a minor impact on the overall energy efficiency of the ammonia synthesis loop, but it significantly affects the required operating temperatures and pressures. Ammonia synthesis reactors are typically designed as multi-bed adiabatic systems where the first beds operate at elevated temperatures (around 500–550 °C) to achieve high reaction rates, while the subsequent beds operate at progressively lower temperatures to maximize conversion [20].

The development of reaction kinetics models and the prediction of the synthesis rate has been

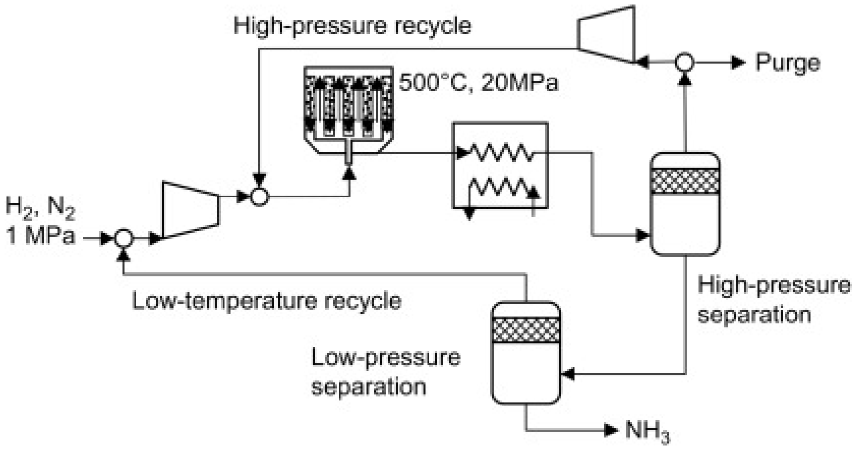


Figure 2.2: Generic schematic of the Haber Bosch process[22]

a focus of research since the beginning of the 20th century. One of the first models was proposed by Temkin and Pyzhev, which assumes the nitrogen adsorption on the catalyst surface as rate-determining. Afterward, numerous modifications to Temkin's work have been presented. Among these, Dyson and Simon introduced a model based on the component activities instead of partial pressures, and have proven to give reliable predictions over iron catalysts at commercial conditions [23]. They expressed the ammonia synthesis rate as:

$$r_{\mathrm{NH_3}} = 2k\left[K_\alpha^2 \alpha_{\mathrm{N_2}} \left(\frac{\alpha_{\mathrm{H_2}}^3}{\alpha_{\mathrm{NH_3}}^2}\right)^a - \left(\frac{\alpha_{\mathrm{NH_3}}^2}{\alpha_{\mathrm{H_2}}^3}\right)^{1-a}\right] \quad [\mathrm{kmol\,m^{-3}\,h^{-1}}] \tag{2.1}$$

Where k, α_i, and K_α represent the rate constant, activity of the respective component, and the equilibrium constant, respectively. The activity can be defined as a function of the fugacity $\alpha_i = \frac{f_i}{f_i^*}$, and the parameter a accounts for the non-uniformity of the catalyst surface, with typical values in the range of 0.25-0.75 [24]. The rate constant is assumed to follow the Arrhenius expression, and the equilibrium constant expresses the relationship between products and reactants of a reaction at equilibrium based on activities [23]:

$$k = k_0 \exp(-E_a/R_{gas}T)) \tag{2.2}$$

$$K_\alpha = \frac{\alpha_{\mathrm{NH_3}}^2}{\alpha_{\mathrm{N_2}}\alpha_{\mathrm{H_2}}^3} \tag{2.3}$$

Where k_0, E_a, R_{gas}, and T are the pre-exponential factor, activation energy, ideal gas constant, and temperature, respectively.

In order to achieve the high inlet pressures to the synthesis reactor with minimal power consumption, a multi-stage compression system is typically used. Because each compression stage raises the temperature and consequently the specific volume of the feed gases, intercoolers are implemented to reduce the required work of the compressors [25].

Pressure Swing Adsorption

As mentioned, the ammonia process is reliant on a nitrogen supply. Nitrogen constitutes 78% of Earth's atmosphere, making air the most abundant nitrogen source. However, for ammonia synthesis, a high-purity nitrogen stream is required, and in ammonia plants, an air separation unit is required, where pressure swing adsorption (PSA) is the most common separation technology. Compressed air (5-12 bar) is first cleaned of oil and moisture using a series of filters before being introduced into one of two adsorption vessels filled with carbon molecular sieves (CMS). At elevated pressures, the CMS selectively adsorbs oxygen and other trace impurities, while nitrogen passes through at the required purity. As one vessel is generating nitrogen, the second is depressurized to release the captured oxygen, which is vented to the atmosphere. This cyclic adsorption–desorption process allows for continuous nitrogen production at purities of up to 99.9995%.[26]. A general diagram of the PSA-process is presented in figure 2.3.

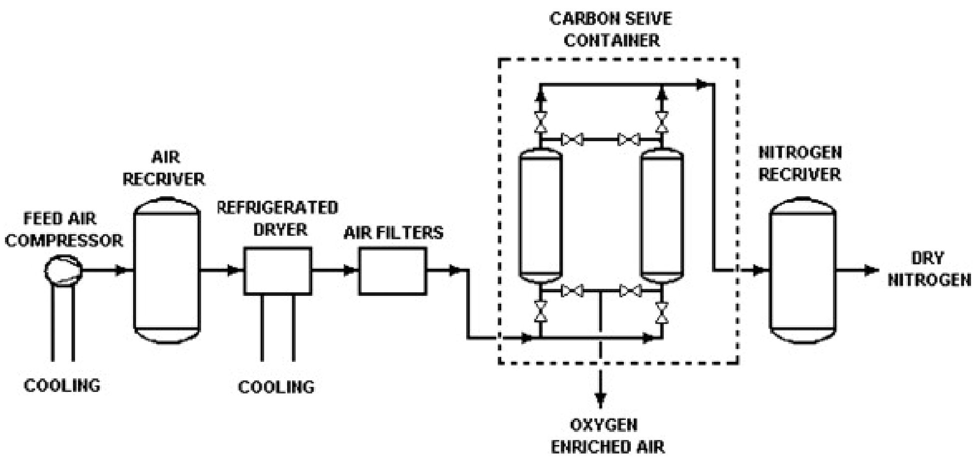


Figure 2.3: Generic schematic of a nitrogen PSA generator[27]

High-Temperature Electrolysis

Large scale production of green ammonia requires novel approaches to non-fossil hydrogen production, where electricity can be used to split water into its components:

$$H_2O \longrightarrow H_2 + \tfrac{1}{2}\,O$$

From a thermodynamic point of view, the process of thermal water splitting becomes more efficient with increasing temperatures, regardless of the specific method [28]. This process is governed by the first law of thermodynamics for a reversible operation, which for an isothermal process is given by:

$$\Delta H_R = T\Delta S_R + \Delta G_R \tag{2.4}$$

The total energy demand for water electrolysis, represented by the enthalpy change of decomposition (ΔH_R), increases slightly with temperature, whereas the Gibbs free energy change, which corresponds to the electrical energy requirement, (ΔG_R), decreases significantly. Consequently, a larger portion of the total energy demand can be supplied as heat, represented by $T\Delta S_R$. This is illustrated in the often cited diagram below.

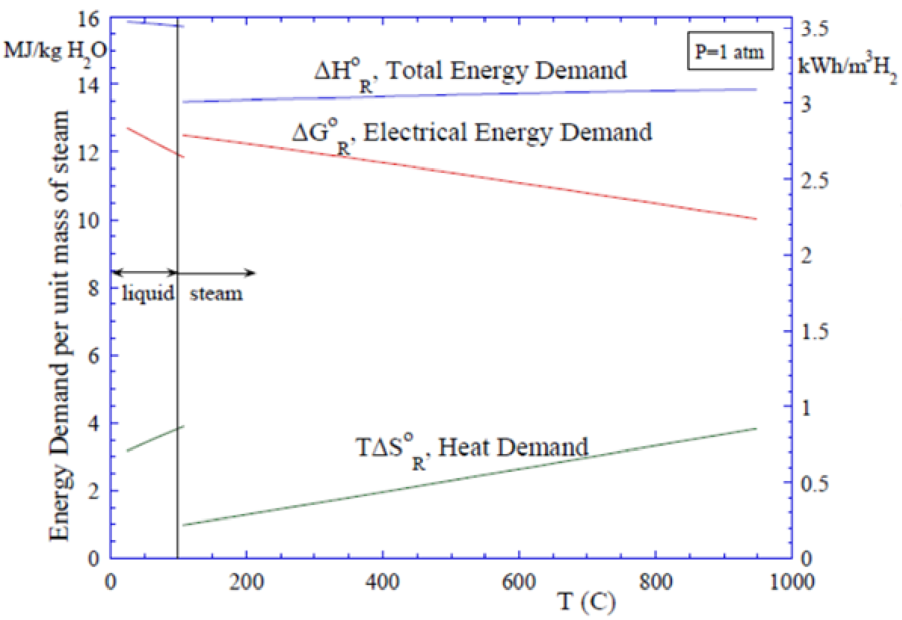


Figure 2.4: Standard-state ideal energy requirements for electrolysis as a function of temperature [28]. The left side y-axis represents the energy demand of splitting the water molecules per unit mass of steam at 1 atm. Right side y-axis displays the energy requirement per cubic meter of produced hydrogen.

Since the conversion of heat to electrical work in heat engines is typically limited to efficiencies of 50% or lower, reducing the electrical work requirement leads to higher overall thermal-to-hydrogen conversion efficiencies. Additionally, the operation of the electrolyzer at high temperatures is also favorable for the reaction kinetics and electrolyte conductivity [28]. As a nuclear reactor

Because of its ability to operate with high temperatures, Solid oxide electrolyzer cells are considered the most efficient electrolysis technology. However, it is important to mention that despite its efficiency advantages, SOECs remain in the development and early commercialization stage, primarily due to challenges related to long-term stability and material degradation. Current reports indicate operating lifetimes in the range of 20,000 to 50,000 hours [29][30].

With typical operating temperatures of 500-850 °C [29], SOECs consist of cells, each having electrically conductive flow channels guiding high-temperature steam, where it is electrochemically split into hydrogen and oxygen. A general diagram of the HTSE-process is presented in figure 2.5.

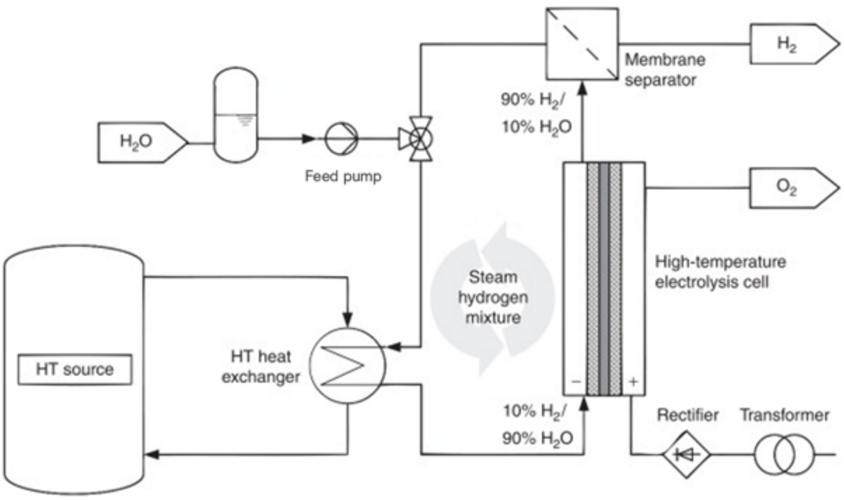


Figure 2.5: Generic schematic of an high-temperature electrolysis system [31]

The electrolysis is an endothermic process, but depending on the operating voltages, the net heat flux in the SOEC stacks may be negative, positive, or zero. The last option is the most desired in regard to thermal stresses on components, as the process becomes isothermal at the thermoneutral voltage. However, there is a trade-off between efficiency and hydrogen production rate when selecting an electrolysis stack operating voltage [28].

The thermal neutral voltage is defined as:

$$V_{tn} = \frac{\Delta H_R}{zF} \tag{2.5}$$

Where $z = 2$ represents the number of electrons transferred per molecule of hydrogen. The number of moles of hydrogen produced at the electrode is linked to the cell current I by Faraday's law of electrolysis [32], which is a function of the current density j and active cell area A_{cell}. The production of hydrogen moles can then be increased by adding cells expressed as [33]:

$$\dot{n}_{\mathrm{H_2}} = \frac{jA_{cell}}{zF} N_{cells} = \frac{I_{cell}}{zF} N_{cells} \tag{2.6}$$

It is often common to apply a single-pass steam utilization (SU) factor that forces a certain level of steam decomposition and determines the required number of electrolysis cells to achieve the conversion:

$$\dot{n}_{\mathrm{H_2}} = SU \cdot \dot{n}_{\mathrm{H_2O}} \tag{2.7}$$

$$N_{cells} = \frac{SU \cdot n_{\mathrm{H_2O}} \cdot zF}{I_{tot}} \tag{2.8}$$

Where I_{tot} is the total current of the stack. The total power input to the stack of cells is defined as:

$$\dot{W}_{stack} = I_{cell} V_{op} \cdot N_{cell} = I_{tot} V_{op} \tag{2.9}$$

Electricity-Ammonia cogeneration

Combining the aforementioned systems and a nuclear power source creates an integrated energy system with several options for thermal integration. In a PWR-type SMR cogeneration configuration, a portion of the steam in the secondary loop is extracted for heat exchange rather than undergoing full expansion in the steam turbines. Depending on the temperature requirements of the integrated process, this thermal energy can be diverted at different stages of the turbine train to match the specific enthalpy needs. A heat exchanger transfers the energy from the high-temperature steam to the required process and can, in some cases, serve as the condenser. Such a cogeneration reactor system is illustrated in the diagram below:

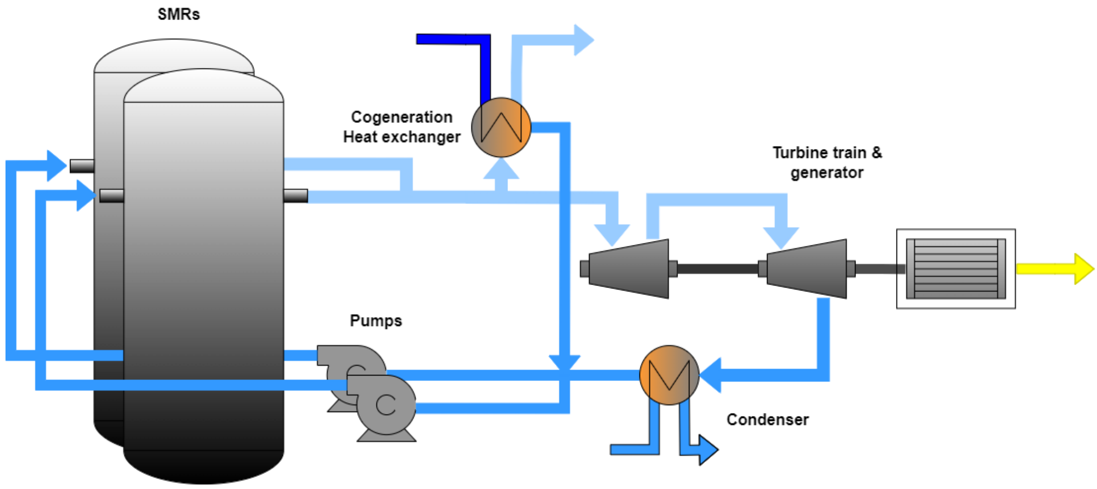


Figure 2.6: Generic diagram of a multi-SMR reactor system with steam extraction for cogeneration.

Although steam extraction alters the thermal balance of the plant and influences the turbomachinery performance, variations in heat demand generally remain within the load-following capabilities of nuclear power plants and do not compromise reactor safety. In some configurations, cogeneration may even improve overall safety by providing an additional heat sink, which is essential for ensuring continuous cooling of the reactor core [3]. The European Utility Requirements (EUR) define the maneuvering performance expected of nuclear power plants in Europe, encompassing a wide range of operating conditions to ensure safe and efficient operation. In particular, the EUR specifies that NPPs must be capable of at least daily load-following operation between 50% and 100% of nominal power, with permissible power ramp rates of approximately 3–5% of rated capacity per minute. Nevertheless, plant designers may offer standard reactor designs that allow operation at lower fractions of rated power, typically down to around 20% [34].

Typical PWR-type SMRs core outlet temperatures are in the range of 280-350 °C [2], meaning that the secondary steam temperatures are also within this range. Consequently, topping heat is therefore required to achieve sufficient SOEC efficiencies, however the majority of the energy consumption in an HTSE system is the steam production...The coupling of an NPP to an industrial process plant requires specific considerations depending on the nature of the industrial processes. Avoiding cross-contamination is important, including preventing radioactive material from migrating from the nuclear reactor to the cogeneration product and preventing potential corrosive chemicals from entering the reactor system from the industrial processes. The nuclear heat energy can be transferred to the industrial process either directly from the primary side through a nuclear-grade heat-exchanging component, or via an intermediate loop (secondary side in PWR). The latter is generally preferred due to safety and operation controllability advantages [3]. Since the HTSE system includes process streams with high temperatures, the heat can be further utilized in downstream processes, creating a cascading utilization of energy, which can improve the overall system efficiency [10].

Flexibility in ammonia production can be achieved to some degree through various mechanisms. One method is to lower the hydrogen availability, such that nitrogen can act as an inert component, allowing continued operation at reduced capacity. However, maintaining a minimum concentration of ammonia in the loop is essential to ensure proper condensation in the refrigeration section. During ramp-down, the specific energy consumption per unit of ammonia produced can rise significantly, though various control strategies have been developed to minimize this increase. For context, a cold start-up of a large-scale ammonia plant typically requires one to two days, meaning that a full shutdown is only justified if the electricity supply is expected to be unavailable for several weeks [20]. Additionally, the profitability of the allocation of thermal energy and continuous production of each commodity within some limits will affect the flexibility.

2.3 System efficiency analysis

When assessing an energy-utilization system, it is essential to examine how energy flows and transforms within it. In this study, the energy conversion processes of the ammonia–hydrogen–electricity cogeneration system are evaluated using the principle of energy conservation. The energy balance of the SMR steam system is:

$$\dot{Q}_{SG} = \dot{Q}_{cogen} + \dot{W}_{turb} - \dot{W}_{p,ss} + \dot{Q}_{cond} \tag{2.10}$$

Where $\dot{Q}_{SG}$ is the thermal energy from the steam generator, $\dot{Q}_{cogen}$ is the energy extracted for cogeneration, $\dot{W}_{turb}$ is the combined electric power output from the turbines, $\dot{W}_{p,ss}$ is the pump work in the steam system, and $\dot{Q}_{cond}$ is the remaining waste heat rejected to the condenser. When the system utilizes the power production for internal components, the net electricity production is

$$\dot{W}_{net} = \dot{W}_{turb} - \dot{W}_{internal} = \dot{W}_{turb} - \dot{W}_{p,tot} - \dot{W}_{comp} - \dot{W}_{heat} - \dot{W}_{refr} - \dot{W}_{stack} \tag{2.11}$$

Where $\dot{W}_{p,tot}$, $\dot{W}_{comp}$, $\dot{W}_{refr}$, $\dot{W}_{heat}$, and $\dot{W}_{stack}$, are the electricity consumption for the total pump work, compressor work, refrigeration, electric heating, and stack electrolysis, respectively.

Energy efficiency is defined as the ratio of the useful energy output to the total energy input. Therefore, the efficiency describing the conversion of thermal energy to electricity is defined as [35]:

$$\eta_{el} = \frac{\dot{W}_{turb} - \dot{W}_{internal}}{\dot{Q}_{Nuclear}} = \frac{\dot{W}_{net}}{\dot{Q}_{Nuclear}} \tag{2.12}$$

A PWR-SMR operates on a Rankine steam cycle, where typical generating efficiencies are between 30 and 35% [2][36]. The energy efficiency of ammonia production is based on the lower heating value of ammonia [37].

$$\eta_{\mathrm{NH_3}} = \frac{\mathrm{LVH}_{\mathrm{NH_3}} \cdot \dot{m}_{\mathrm{NH_3}}}{\dot{Q}_{Nuclear}} \tag{2.13}$$

where $\mathrm{LVH}_{\mathrm{NH_3}}$, and $\dot{m}_{\mathrm{NH_3}}$ are the lower heating value of ammonia (18,6 MJ/kg) [38], and the mass flow of the final ammonia product, respectively. Because of cogeneration, an overall ammonia and electricity utilization efficiency can be defined, accounting for all the useful energy output.

$$\eta_{cogen} = \frac{\mathrm{LVH}_{\mathrm{NH_3}} \cdot \dot{m}_{\mathrm{NH_3}} + \dot{W}_{net}}{\dot{Q}_{Nuclear}} = \eta_{\mathrm{NH_3}} + \eta_{el} \tag{2.14}$$

Heat exchangers are utilized for fluid heat transfer across the system. Dimensioning of heat exchangers is commonly described by [39]:

$$\dot{Q}_{ex} = UA_{hx}\Delta T_{LMTD} \tag{2.15}$$

Where U is the overall heat transfer coefficient, A_{hx} is the heat exchanger area, ΔT_{LMTD} is the log mean temperature difference.

2.4 Techno-economic analysis

In this section, the theory behind the techno-economic analysis is presented. At the early stages of project development, a cost-of-manufacturing estimate provides vital insights into economic viability and highlights key targets for cost reduction. While preliminary estimates are limited by uncertainty regarding the inputs, these unknowns can be treated as variables within realistic ranges to identify significant cost drivers. Researchers often underestimate their ability to establish these ranges and overestimate the accuracy required for a functional prediction. However, a complete Techno-Economic Analysis (TEA) must eventually balance these costs against market value and revenue potential [40]. While this study focuses specifically on the cost side of the ledger, it is recognized that marketplace value often has a more substantial impact on overall project economics than manufacturing costs alone.

The revenues of electricity and ammonia production also accompany the production cost. Therefore, it is essential to describe the operating cost drivers of the system. In this work, all monetary values are presented in 2025 Euro-based values and escalated from the respective price years using price indexes [21]:

$$C_2 = C_1 \frac{I_2}{I_1} \tag{2.16}$$

If data on costs and prices used are in USD-based values, they have been converted to euros using the average value (1.12) for the past ten years from the European Central Bank [41].

2.4.1 SMR cost

The economics of new nuclear power projects are mostly influenced by their capital cost, where the O&M costs have historically been relatively low [42]. The assessment of the future costs of new energy technologies, such as SMRs, is a complex task due to a multitude of uncertainties and variables. Therefore, it is subject to various approaches, and different methods are applied in the literature. However, the costs are often divided into either the overnight capital cost (OCC) or the capital cost, and the operating cost (OPEX). The first terms refer to the fixed costs of engineering, procurement, construction, infrastructure, and various contingencies. While the OCC and the capital cost are sometimes used interchangeably, they differ in that the capital cost also accounts for the financing costs during construction [43]. The OPEX accounts for the expenses related to fuel, plant operation, maintenance, and the allocated funds required for decommissioning, as well as the management and disposal of spent fuel and radioactive wastes [42]. Furthermore, the OPEX can be divided into fixed and variable costs.

The Idaho National Laboratory has carried out a comprehensive statistical cost assessment of SMRs and large reactors in the US [43]. Based on a sensitivity analysis, they found limited variations in OCC between SMRs and large reactors, while microreactors appeared to be more expensive. Their cost estimates represent a nuclear construction project between first-of-a-kind and Nth-of-a-kind (BOAK), and are reported in 2019 USD across three quartiles, indicating how much of the dataset is above or below the value in percentage. This includes the OCC, OPEX, a Multi-unit OCC Exponent (MUE), and an OPEX multi-unit multiplier (MOM) to adjust for cost reduction when constructing multiple units. Additionally, they define a FOAK premium and learning rate to adjust for the higher cost associated with a FOAK build and the cost reduction associated with serial production, respectively. Table 2.1 presents the cost parameters escalated to 2025 values using the same producer price index as in the INL-report for the respective year [44].

Table 2.1: Summary of identified BOAK cost estimation values and correction factors for large reactors and SMRs converted to 2025 values and Euros [43].

Parameter	**Lower (25%)**	**Median (50%)**	**Upper (75%)**	σ
OCC	€6,133/kWe	€9,199/kWe	€10,732/kWe	€5,826/kWe
OPEX	€23/MWh	€38/MWh	€54/MWh	€23/MWh
FOAK premium	1.3	1.6	2.1	-
Learning rate (LR)	5%	10%	15%	-
Multi-Unit OCC exponent	0.8	0.825	0.850	-
Multi-Unit OPEX Multiplier	0.5	0.624	0.7	-

In this study, the economic analysis focuses on the operational expenditure, and the relevant cost parameters are therefore the SMR-OPEX and MOM. The MOM prefactor reduces the initial value because the cost per unit power produced decreases for every additional SMR unit. The factor is used as follows:

$$OPEX_{multi-unit} = OPEX_{1unit} \times MOM \tag{2.17}$$

2.4.2 Chemical plant equipment and operation cost

Preliminary equipment cost estimation methods for new chemical plants are presented by Turton et al. [21] and can be applied to estimate the remaining system costs. The cost of the steam turbine cycle is already accounted for in the estimations by INL, except for the cogeneration heat exchanger. The proposed modular costing technique presents a general bottom-up cost estimation and relates all costs to the purchase cost of equipment evaluated at specified base conditions. Deviations from these base conditions arising from factors such as equipment type, operating pressure, and construction materials are accounted for through the use of appropriate multiplying factors. All prices mentioned in the book are stated in 2001 values with a Chemical Engineering Plant Cost Index (CEPCI) of 397.

Depending on whether the project involves the construction of a completely new facility, or making expansions or alterations to an existing facility, the fixed capital cost is estimated by the grass-roots (GR) cost or total module (TM) cost:

$$C_{TM} = \sum_{i=1}^{n} C_{TM,i} = 1.18 \sum_{i=1}^{n} C_{BM,i} \tag{2.18}$$

$$C_{GR} = C_{TM} + 0.5 \sum_{i=1}^{n} C^{\circ}_{BM,i} \tag{2.19}$$

Where n represents the number of equipment. C°_{BM} and C_{BM} refer to the bare module cost at base and non-bane conditions. The prefactor in equation 2.18 accounts for the contingency cost and fees, taken as 18% of the bare module cost. The prefactor in equation 2.19 accounts for the auxiliary facility costs taken as 50% of the bare module cost at base conditions.

Operating conditions different from base conditions might require different equipment materials, which increases cost. This is covered in the bare module cost C_{BM}, which depends on the bare module cost factor F_{BM}. The correlations for the bare module cost are different for different types of equipment, and are listed in table 2.2.

The constants B_1 and the material cost factor F_M for heat exchangers, vessels, and pumps depend on the selected type of equipment and material.

The purchased equipment cost at base conditions, which is carbon steel construction at near

Table 2.2: Bare Module Cost Equations for Selected Equipment Types

Equipment Type	Equation for Bare Module Cost	Note
Compressors	$C_{BM} = C_p^0 F_{BM}$	
Drives for compressors	$C_{BM} = C_p^0 F_{BM}$	
Demister pad	$C_{BM} = C_p^0 N F_{BM}$	N is number of pads
Heat exchangers, vessels, pumps	$C_{BM} = C_p^0 (B_1 + B_2 F_M F_P)$	B_1, B_2 are constants

ambient pressures, is described as:

$$\log_{10} C_p^{\circ} = K_1 + K_2 \log_{10}(A) + K_3[\log_{10}(A)]^2 \tag{2.20}$$

Where A and K_i are equipment capacity/size and type parameters, respectively. The relevant parameter values for this work are presented in table **??**

Table 2.3: Purchase cost Correlation Parameters for Equipment Types

Equipment Type	K1	K2	K3	Capacity Units	Min Size	Max Size
Heat Exchanger (Floating head)	4.8306	-0.8509	0.3187	HX area (m^2)	10	1000
Compressor (Cnetrifugal)	2.2987	1.3604	-0.1027	Fluid Power (kW)	450	3000
Enclosed Electric Drive	1.9560	1.7142	-0.2282	Shaft Power (kW)	75	2600
Process vessel (Vertical)	3.4974	0.4485	0.1074	Volume (m^3)	0.3	520
Demister Pad	3.2353	0.4838	0.3434	Area (m^2)	0.7	10.5
Pump (centrifugal)	3.3892	0.0536	0.1538	Shaft Power (kW)	1	300

The operation of equipment above ambient pressure increases the cost of the equipment, which is covered by the pressure factor F_p. For a process vessel, the pressure factor depends on the diameter D [m], and operating pressure P_{vessel} in barg, the factor is expressed as:

$$F_{p,vessel} = \frac{\frac{(P_{vessel}+1)D}{2[850-0.6(P_{vessel}+1)]} + 0.00315}{t_{min}} \tag{2.21}$$

This corresponds to a vessel at base material conditions with a maximum allowable stress for carbon steel at 944 bar, and a minimum allowable wall thickness of $t_{min} = 0.0063$m. For other process equipment, the pressure factor is based on equipment-specific constants C_i, and is given by:

$$\log_{10} F_p^{\circ} = C_1 + C_2 \log_{10}(P) + C_3[\log_{10}(P)]^2 \tag{2.22}$$

Where P is the design pressure in barg. The values of the constants in equation 2.22 relevant for this work, are listed in table E.1, in appendix D. Some equipment does not have pressure ratings and therefore have the constants equal to zero.

The operating cost of the chemical plant is obtained by combining the following cost components:

- Fixed capital investment (FCI Taken as C_{GR})
- Cost of operating Labour (C_{OL})
- Cost of Raw Material (C_{RM})
- Cost of Utilities (C_{UT})
- Cost of Waste treatment (C_{WT})

The relation between these defines the operating cost expressed in dollars per unit time, as:

$$OPEX_{\mathrm{NH_3}} = 0.280\text{FCI} + 2.73C_{OL} + 1.23(C_{RM} + C_{UT} + C_{WT}) \tag{2.23}$$

This includes depreciation taken as 10% of the FCI. The operating labor costs are based on the hourly wage and depend on the required number of operators. This is given by:

$$N_{operator} = (6.29 + 31.P_{PS}^2 + 0.23N_{NP})^{0.5} \tag{2.24}$$

Where P_{PS} is the number of processing steps involving particulate solids, and $N_{NP} = \sum n_{Equipment}$, is the number of non-particulate processing steps accounted for by summing the number of processing equipment (excluding pumps and vessels). In this work, P_{PS} is set to zero. The equation does not include any support or supervisory staff.

2.4.3 European power prices

A generator in the power market earns revenues based on the received wholesale electricity price [45]. The European Network of Transmission System Operators for Electricity (ENTSO-E), and provides wholesale electricity price data for most of the EU/EEA countries. Table 2.4 presents the median electricity prices in 2025 for selected European markets, based on nominal hourly wholesale price data from EMBER. Their data are sourced from ENTSO-E, EMR for the UK, and SEMOpx for Ireland.

Table 2.4: Median electricity prices between 2025-01-01 to 2026-01-01 per country in €/MWh rounded to the nearest hundredth, based on power price data from EMBER [46].

Country	Value
Austria	104.97
Belgium	82.40
Bulgaria	112.60
Croatia	109.41
Czechia	100.20
Denmark	81.65
Estonia	82.96
Finland	42.13
France	61.40
Germany	90.97
Greece	106.19
Hungary	113.76
Ireland	114.38
Italy	121.65
Latvia	85.74

Country	Value
Lithuania	85.45
Norway	44.86
Netherlands	120.27
North Macedonia	108.71
Poland	108.93
Portugal	66.16
Romania	108.16
Serbia	118.00
Slovakia	102.64
Slovenia	104.65
Spain	65.29
Sweden	42.64
Switzerland	98.10
United Kingdom	94.36

2.4.4 European Ammonia prices

The ammonia selling price is strongly linked to the price of natural gas, where the Title Transfer Facility (TTF) in the Netherlands is one of the main pricing points for natural gas in North-West Europe [47]. Based on data from [48], historical blue ammonia (fossil-based) CFR prices between 1986-2024 in North-West Europe are shown in figure 2.7.

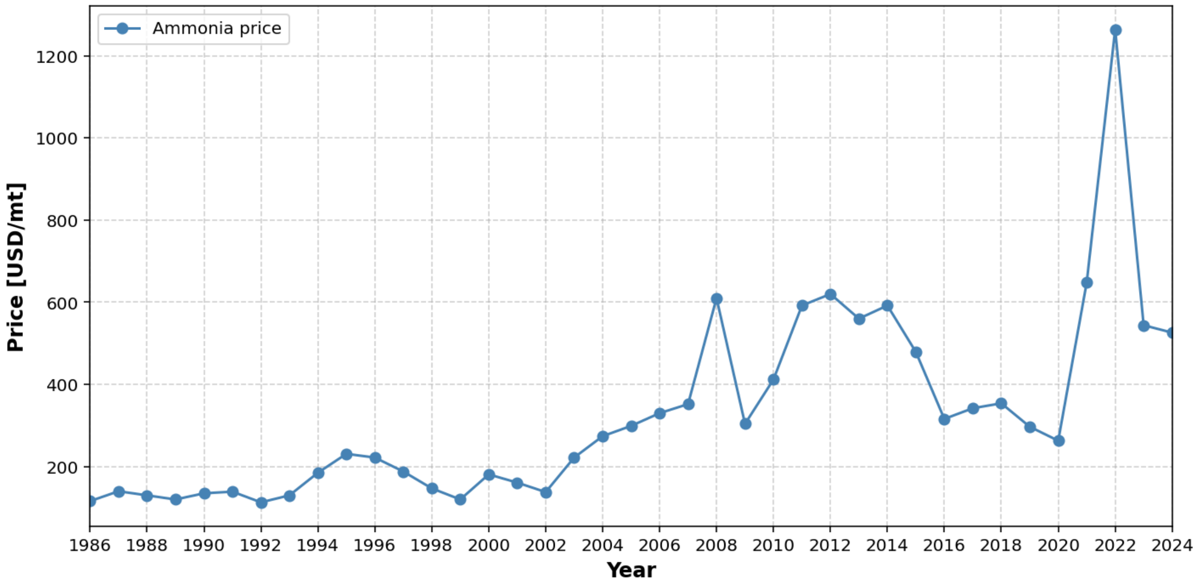


Figure 2.7: Ammonia CFR prices in NW Europe

Following a sharp escalation in 2021, ammonia prices peaked in 2022 and have maintained an upward trajectory in subsequent years. Market intelligence from various commodity providers indicates that European blue ammonia spot prices currently range between 530 – 690 USD/t (about 590 – 770 €/t) as of 2025, carbon-free green ammonia produced is priced upwards of 900 USD/t (1,008 €/t) [49][50][51].

3 Methodology

This section describes the development of the models that serve as the foundation for this study. The aim of the research is to investigate the techno-economic potential of an SMR-driven power and ammonia cogeneration plant, via HTSE using SOECs, in the European power trading market. This is conducted through a combination of investigating the thermodynamic steady-state operation of the system and its operating costs. The first part considers parameters including the production capacities, equipment operating ranges, efficiencies, and allocation of thermal energy. The latter part of the analysis is primarily tied to costs and commodity price data. This considers parameters such as equipment size and material of manufacturing, operational expenditures, and electricity and ammonia price time series. It is therefore necessary to perform the thermodynamic and process calculations first to acquire the production volumes and rated capacities of components. When the production capacities and OPEX are acquired, the revenues can be determined by using the market prices of each commodity.

3.1 DWSIM: Chemical Process Simulator

To model the integrated energy system on a plant scale, the open-source chemical process simulator DWSIM was utilized. DWSIM is a software tool for modeling, simulating, and optimizing steady-state and dynamic chemical processes developed by the chemical engineer Daniel Wagner Oliviera de Medeiros. The tool has been utilized and cited in numerous scientific studies and articles, and has been evaluated to perform comparably to other commercial simulators [52][53]. The developed model captures the core components governing the material and energy streams, spanning from steam generation to ammonia synthesis from thermodynamic, electrochemical, and reaction kinetic perspectives. However, it excludes peripheral Balance of Plant (BOP) components such as filtration, pressure regulation, and thermal balancing systems—required for actual operation. Integrating these components would significantly increase the model's complexity beyond the scope of this preliminary analysis

In regard to the mentioned utility requirements and flexibility in ammonia production, the specification of how to manage the flexible operation and meet regulatory requirements is out of the scope of this work. Furthermore, it was assumed that the power generated by the turbines could supply the required electric power for all equipment associated with the HTSE and ammonia synthesis. This includes power for resistance heating, SOEC, separation, pumps, compressors, blowers, and refrigeration.

3.1.1 Secondary steam system model

The secondary steam system model was derived from the BOP configuration with a high-temperature extraction point, developed in the work of SIMONINI et al. [54]. A few additional configurations were adopted from [55], based on the same BOP. The BOP architecture is part of the TANDEM project for the European Small Modular Reactor (E-SMR) with a thermal power of 540 MW. The E-SMR is a conceptual SMR design introduced within the Euratom ELSMOR project, which was initiated to examine key safety characteristics of light-water SMRs, with particular emphasis on licensing challenges. The TANDEM project was a European initiative funded under the EURATOM program, aimed at to conduct evaluating the nuclear safety, techno-economic performance, and operational feasibility of Hybrid Energy Systems (HES) integrating SMRs [54][56]. The main parameters of the reactor are reported in the following table:

Parameter	**Value**
Thermal Power	540 MW
Core inlet temperature	300°C
Core outlet temperature	324.5°C
Nominal pressure	150 bar
Nominal coolant flow rate	3700 kg/s

Table 3.1: E-SMR parameters.

The steam loop is simulated as a Rankine cycle and comprises a high, intermediate, and low-pressure turbine (HPT, IPT, LPT), a moisture separator, two pumps, and three reheaters. As this work only considers the steady state solutions, the control valves, as well as the feed water tank, were deemed unnecessary to implement. Hence, the tank was simply replaced with a stream mixer. The turbine objects in DWSIM do not allow for more than one outlet stream from the turbines, and a stream splitter was used instead for steam diversion to the preheaters. The flowsheet of the steam system with the extraction point is presented in figure 3.1.

The steam generator evaporates the pressurized water. Then a portion of the steam is diverted for cogeneration heat exchange, particularly suitable for the production of industrial process

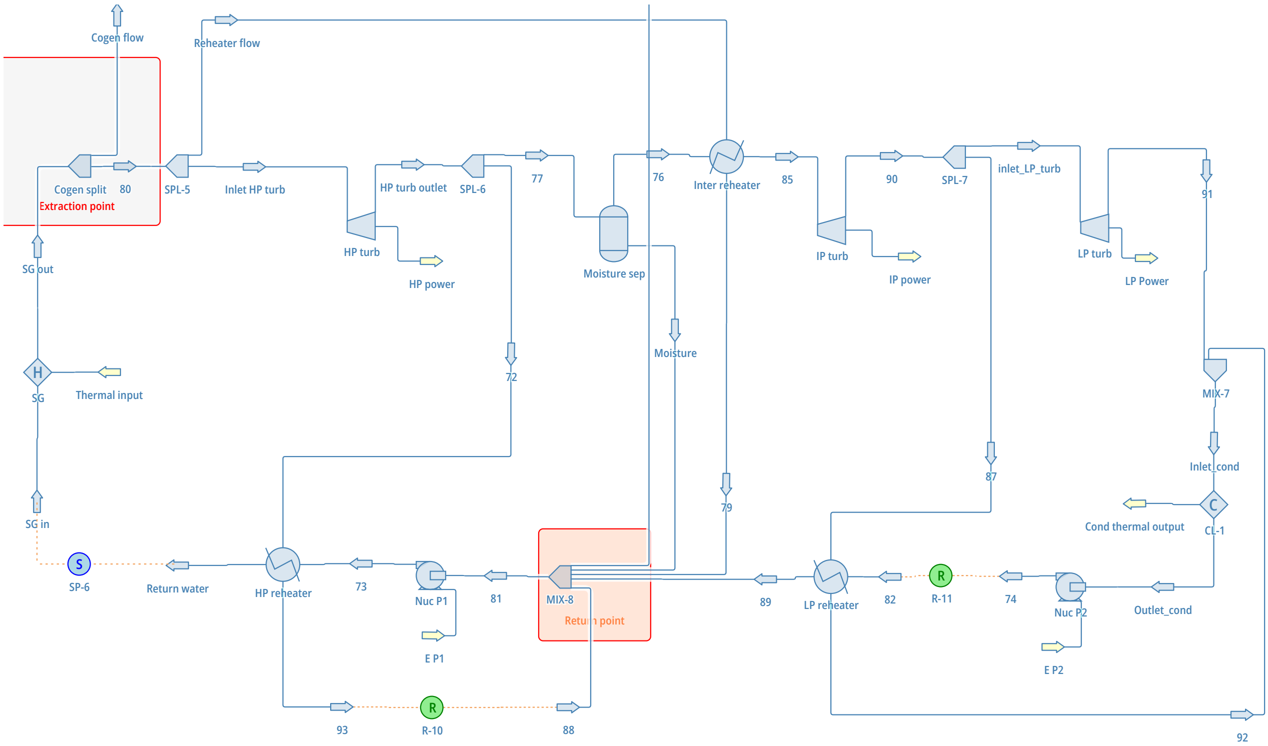


Figure 3.1: Flowsheet of nuclear reactor secondary steam loop designed in DWSIM. Steam extraction and return point are highlighted in red colored rectangles.

steam, with the return point to the live steam before the HPT [55]. The stream is then routed to the first reheater and expanded in the HPT. The flow exiting the HPT is partially routed to the HP-reheater to increase the feedwater temperature entering the SG. The remaining part of the flow enters the IPT after passing through a moisture separator and the following intermediate reheater. The liquid drained by the moisture separator is recycled, together with the flow at the intermediate reheater's hot side outlet. A steam extraction from the LPT drives a low-pressure reheater. The condensate from the high-pressure and low-pressure reheaters is recycled to the first pump and condenser, respectively.

Because not all of the design parameter data from the reference steam cycle were available to the author, some assumptions were made to acquire the required inputs to the flowsheet objects in DWSIM. Hence, the turbine efficiencies were adopted from [55], and corrected for wetness losses through Baumann's rule as described in the reference paper:

$$\eta_{iso} = \eta_{iso}^{nom}(1 - (1 - x_m)C_B) \tag{3.1}$$

Where η_{iso}, and η_{iso}^{nom} are the nominal, and non-nominal isentropic efficiency. x_m is the average steam quality and C_B is the Baumann coefficient set equal to 1 [55]. Furthermore, the outlet pressure of the first pump (Nuc P1) was assumed based on the heat exchanger pressure drop and the SG inlet pressure in the reference literature, while the outlet pressure of the second pump (Nuc P2) is based on the secondary loop design pressure. The design parameters are listed in table 3.2, and the steam tables IAPWS-IF97 is used as the fluid package.

Table 3.2: Design parameters of the secondary steam system.

Parameter	**Value**	**Unit**	**Reference**
SG outlet temperature	300	°C	[56]
SG mass flow	239.64	kg/s	[55]
SG inlet pressure	45.4	bar	[54]
SG outlet pressure	44.4	bar	[54]
HPT outlet pressure	7.547	bar	[54]
IPT outlet pressure	0.801	bar	[54]
LPT outlet pressure	0.07	bar	[54]
HX pressure drop	0.4	bar	[54]
HP reheater heat load	47.52	MW	[54]
LP reheater heat load	37.72	MW	[54]
Condenser outlet vapor fraction	0	-	*
HPT efficiency	0.9	-	[55]
IPT efficiency	0.89	-	[55]
LPT efficiency	0.89	-	[55]
Nuc P1 outlet pressure	45.8	bar	*
Nuc P2 outlet pressure	45	bar	*
Reheater hot-side flow	21.32	kg/s	[54]
HP Reheater hot-side flow	24.51	kg/s	[54]
LP Reheater hot-side flow	16.41	kg/s	[54]

*Assumed values based on information in references

As this research focuses on a steady-state analysis, a detailed investigation of the underlying thermodynamics in the steam cycle lies beyond the current scope. Therefore, the validity of the steam cycle model was established by adopting the outputs of the CYCLOP model in the reference literature as fixed inputs and verifying the consistency of the converged parameters. By utilizing the validated CYCLOP model as a benchmark, the study ensures thermodynamic consistency while allowing the focus to remain on the higher-level integration of the SMR with the ammonia production system. The simulation results of the secondary steam system without steam extraction are presented in section 4.1.

3.1.2 Chemical plant model

The chemical plant model and design, consisting of the electrolysis and ammonia synthesis systems, was adopted from M. Schiedeck et al. [57], with a design production capacity of 1356 tdp of ammonia at full load. However, their work focused on the optimization of the energy integration of the system and also included the modelling of a refrigeration unit, both being out of the scope of this work. Additionally, they refer to the work by Fahra et al. [58] for the Haber-Bosch process. Therefore, in this work, the SOEC system, except the BOP configuration, was modelled after the former-mentioned author, while the Haber-Bosch process was taken from the latter.

High-temperature steam electrolysis model

The HTSE system consists of the electrolysis stacks and the BOP-components heat exchanger, topping heater, recirculation blowers, and hydrogen dryer. The SOEC system is assumed to operate at thermoneutral conditions, with a stack inlet pressure and temperature of 1.05 bar and 750 °C, respectively, taken directly from M. Schiedeck et al. This corresponds to a thermoneutral voltage of $V_{op} = 1.285V$. The inlet feed water is pumped to a pressure of 1.35 bar (to overcome pressure drops in the HTSE system) and evaporated in the cogeneration heat exchanger before the stream is mixed with the unreacted recycle steam and heated through a heat exchanger, recovering heat from the stack outlet. An electric topping heater ensures the required steam inlet temperature to the stacks. The anode side is operated without a sweep gas, where oxygen is allowed to evolve undiluted. This offers higher system efficiencies, but requires special safety measures to handle the hot pure oxygen. The cathode outlet off-gas is dried by a compound separator, which is assumed to be a PSA unit. The hydrogen separator factor was set to 86.2%, corresponding to an approximate 90/10 molar fraction of steam and hydrogen in the stack inlet. A heat exchanger between the steam feed and the anode outlet oxygen was implemented to increase the heat utilization. A blower on each side maintains the operating pressure. The Peng-Robinson fluid property package was utilized in the model. 3.2. The flowsheet of the HTSE system modeled in DWSIM is presented below:

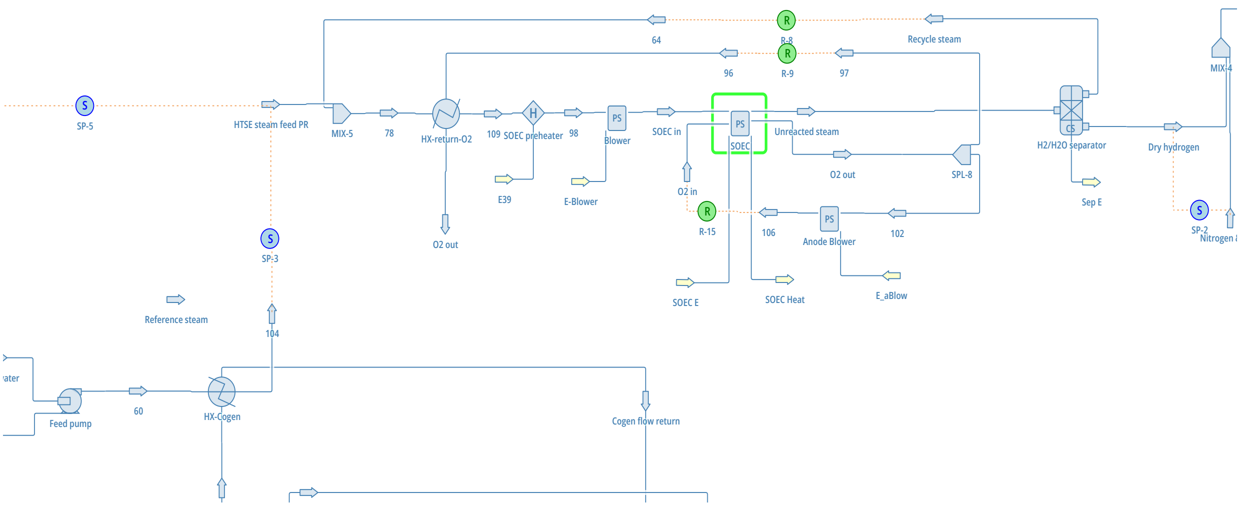


Figure 3.2: Flowsheet of HTSE system designed in DWSIM. SOEC component is highlighted by green rectangle. Feedwater is pumped into the system from the left, and dry hydrogen leaves the system to the right

The cell model was developed using a "Python script" user model in DWSIM. The operating cell voltage in equation 3.2 is described by the reversible cell voltage calculated through the Nernst equation, and the various losses in the cells.

$$V_{op} = V_{rev} - jR_{\Omega} - \eta_{act,\mathrm{H_2}} - \eta_{act,\mathrm{O_2}} \tag{3.2}$$

$$V_{rev} = \frac{\Delta G_r}{zF} + \frac{RT}{zF} \ln \left[\left(\frac{p_{\mathrm{H_2},TPB}}{p_{\mathrm{H_2O},TPB}} \right) \left(\frac{p_{\mathrm{O_2},TPB}}{p_0} \right)^{1/2} \right] \tag{3.3}$$

The Gibbs free energy of the reversible cell is calculated (in positive value) using the Shomate equations and coefficients for the respective gas components from NIST data [59]. The thermodynamic equilibrium, as defined by the Nernst equation, is a function of the reactant and product concentrations at the triple-phase boundary (TPB), where the ion-conducting electrolyte, electron-conducting phase, and gas-filled pores intersect. The ohmic resistance losses are expressed through an Arrhenius approach as such:

$$R_{\Omega} = \frac{T}{B_{\Omega}} \exp \left(\frac{E_{act,\Omega}}{RT} \right) \tag{3.4}$$

Where the material-specific constant B_Ω and activation energy $E_{act,\Omega}$ are fitted parameters. The Butler-Volmer equation 3.6, 3.8 is used to determine the activation overpotentials $\eta_{\mathrm{H_2/O_2}}$.

$$j = j_{0,\mathrm{H_2}} \left[\exp\left(\frac{\alpha_{\mathrm{H_2}} zF \cdot \eta_{act,\mathrm{H_2}}}{RT} \right) - \exp\left(-\frac{(1-\alpha_{\mathrm{H_2}}) zF \cdot \eta_{act,\mathrm{H_2}}}{RT} \right) \right] \tag{3.5}$$

$$j_{0,\mathrm{H_2}} = \gamma_{\mathrm{H_2}} T \left(\frac{p_{\mathrm{H_2},TPB}}{p_0} \right)^a \left(\frac{p_{\mathrm{H_2O},TPB}}{p_0} \right)^b \exp\left(\frac{-E_{act,\mathrm{H_2}}}{RT} \right) \tag{3.6}$$

$$j = j_{0,\mathrm{O_2}} \left[\exp\left(\frac{\alpha_{\mathrm{O_2}} zF \cdot \eta_{act,\mathrm{O_2}}}{RT} \right) - \exp\left(-\frac{(1-\alpha_{\mathrm{H_2}}) zF \cdot \eta_{act,\mathrm{O_2}}}{RT} \right) \right] \tag{3.7}$$

$$j_{0,\mathrm{O_2}} = \gamma_{\mathrm{O_2}} T \left(\frac{p_{\mathrm{O_2},TPB}}{p_0} \right)^m \exp\left(\frac{-E_{act,\mathrm{O_2}}}{RT} \right) \tag{3.8}$$

M. Schiedeck et al. [57], accounted for the diffusion transport in the porous electrodes, and its influence on the partial pressures at the TPBs through the dusty-gas model. However, a simpler approach was taken in this work by using the Fickian model instead [60]. The TPB partial pressures are then obtained by:

$$p_{i,TPB} = p_i - \frac{\delta_{ca/an} RT}{zF D_{i,eff}} j \tag{3.9}$$

Where $\delta_{ca/an}$ is the thickness of the cathode or anode, and $D_{i,eff}$ is the effective diffusion coefficient of each species [61]. The latter coefficient can be expressed by the Bosanquet formula, which includes the same molecular $D_{ij,M}$ and Knudsen $D_{i,kn}$ diffusion as in DGM:

$$\frac{1}{D_{i,eff}} = \frac{\tau}{\varepsilon} \left(\frac{1}{D_{ij,M}} + \frac{1}{D_{i,kn}} \right) \tag{3.10}$$

The power requirement for the recirculating blowers is modeled using the simple relation [62]:

$$E_b = \frac{\Delta P_{i,o} \cdot \dot{V}}{\eta_b} \tag{3.11}$$

Where $\Delta P_{i,o}$, $\dot{V}$, and η_b are the pressure increase, volumetric flow rate of the gas, and blower efficiency, respectively.

The Python script for the SOEC model is attached in appendix A along with model parameters. The Butler-Volmer equations were solved numerically, using the Newton-Raphson method, and a steam utilization of 70% was assumed [63]. A validation of the SOEC model is presented in section 4.1.

Pressure Swing Adsorption model

The nitrogen gas from PSA with a small fraction of argon was modelled as a black box, defined by a single material stream in DWSIM that mixes with the dry hydrogen from the HTSE system. The stream properties were assumed with an inlet pressure of 5 bar and a temperature of 35 °C [20]. The inlet molar flow rate was specified by a specification block connected to the hydrogen stream from the HTSE system, using a target function of $Y(x) = x_{\mathrm{H_2}} \cdot 0.336$, corresponding to a H2/N2 = 3 when the nitrogen stream contains 1% inert. The energy requirement from the PSA unit was considered negligible in this work.

Haber-Bosch model

The ammonia reactor design and modelling of reaction kinetics were adopted from Fahra et al. [58]. The configuration follows the Topsoe S-300 converter design, and comprises three adiabatic beds succeeded by three heat exchangers. The upper two heat exchangers enable indirect intercooling (IC) to improve reaction rates, while the lower internal feed-effluent heat exchanger (iFEHE) is used for heat recovery. The feed is preheated to 140 °C, and split between the three heat exchangers (IC 1, IC 2, iFEHE) and the cold shot (gas bypass). Fahra et al. incorporated a steam production system using the ammonia reactor effluent, which was omitted in this work. The flowsheet of the HB model developed in DWSIM is presented in figure 3.3.

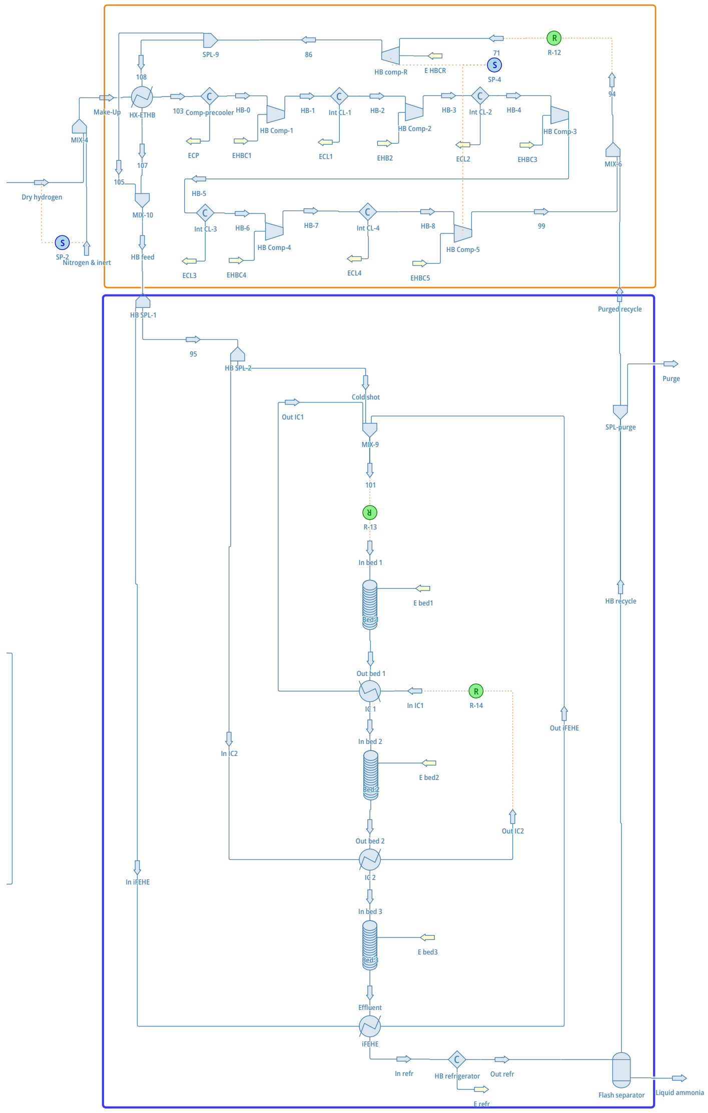


Figure 3.3: Flowsheet of the multi-stage compression (orange rectangel) and Haber Bosch process (blue rectangel). The nitrogen feedstock from PSA is indicated in the left most vertical stream.

The thermodynamics is based on the Peng-Robinson (PR) equation of state (EOS) for all components except the separator, where they utilized the Peng-Robinson-Stryjek-Vera (PRSV) for better vapor-liquid equilibrium predictions. The reaction rate expression is based on the one by Dyson & Simon with $a = 0.5$, yielding:

$$r_{\mathrm{NH_3}} = k \left(K_f(T, p_0)^2 f_{\mathrm{N_2}} \frac{f_{\mathrm{H_2}}^{1.5}}{f_{\mathrm{NH_3}} (p_0)^2} - \frac{f_{\mathrm{NH_3}}}{f_{\mathrm{H_2}}^{1.5}} \right) \tag{3.12}$$

To be compatible with the PR-EOS at a reference pressure of $p_0 = 1$ bar, Fahra et al. modified both the rate constant and the expression by Gillespie and Beattie for the equilibrium constant.

The leading factor 2 was therefore incorporated into the rate constant, and the proposed values for the pre-exponential factor and activation energy are $k_0 = 1.182979e15\,\mathrm{kmol\,bar^{0.5}\,m^{-3}\,h^{-1}}$, and $E_a = 168110.6$ J mol^{-1}. The equilibrium constant is defined as:

$$\ln(K_f(T, p_0)) = 3.963014 + \frac{4.554.634427[\mathrm{K}]}{T} - 2.315092 \cdot \ln\left(\frac{T}{1[\mathrm{K}]}\right) \tag{3.13}$$

The catalyst beds were implemented as adiabatic Plug-Flow-Reactors (PFR) objects in DWSIM, and the governing equations were implemented by selecting a kinetic reaction connected to the internal Python script in DWSIM. Catalyst bed volumes and split factors of the stream splitters were adopted unchanged from the result of the design optimization by Fahra et al. However, since the stream splitter object in DWSIM only allowed for a two-way split, two stream splitters in series were implemented, where the feed stream was first split to the iFEHE by the same factor 9.9%. This requires the next split factors for the ICs and the cold shot to be adjusted by $\xi_{new} = \xi_{original}/(1 - 0.099)$, to represent the same stream conditions. At full load, the reactor is operated at a pressure of 150 bar with a nominal production rate of 1356 tons per day (tpd), using a stoichiometric composition of $x_{\mathrm{H}_2}/x_{\mathrm{N}_2} = 3$ in front of the first bed. The pressure drop within each bed and heat exchangers is assumed to be 0.6 and 0.3 (each side) bar, respectively. The reactor effluent is cooled to -5°C by an ammonia chiller for separation, with a pressure drop of 3 bar. This unit was assumed to have a coefficient of performance of 2.5. The unreacted part of the effluent is recycled and combined with the feed stream before recompression and preheating. A validation of the reactor model is presented in subsection 4.1.

Fahra et al. assumed a purge-free HB-system, requiring low impurity levels in the feed gases. However, in this work, a purge stream of 1% of the recycle stream was implemented after the reactor model validation to account for an argon impurity of 1% from the nitrogen PSA unit.

A multi-stage compression system similar to the one in [58] with intercoolers was implemented to compress the make-up hydrogen, nitrogen, and inert gas mix. It comprises five compression stages, assuming centrifugal compressors with electric drives. The hot discharge of each compression is cooled down to 40 ° C, and the compression ratios were determined using the correlations in appendix B, corresponding to an equal compression ratio of $r_{comp} = 2.29$, which also accounts for an assumed intercooler pressure drop. The recycle compressor was defined with an outlet pressure of 150.2 bar to overcome pressure losses in the HB-system. Additionally, a heat exchanger was added for heat recuperation between the hydrogen and the recycle ammonia reactor effluent, and a bypass across the heat exchanger ensures inlet temperature control.

Table 3.3: Summary of design parameters in the Haber Bosch model

Object	**Parameter**	**Value**	**Ref**
SPL-1	ξ_{iFEHE} (%)	9.9	[58]
SPL-2	ξ_{IC} (%)	88.9	*
SPL-2	ξ_{coldshot}(%)	11.1	*
Bed$_{1,2,3}$	ΔP (bar)	0.6	[58]
Bed$_{1,2,3}$	D_{bed} (m)	2.0	¤
HX$_i$	ΔP_{hx} (bar)	0.3	[58]
iFEHE	η (%)	65	[58]
HB-refr	ΔP_{refr} (bar)	3.0	[58]
HB-refr	T_{out} (°C)	-5	[58]
HB-refr	COP	2.5	¤
HB Comp$_i$	r_{comp} (-)	2.29	¤
HB Comp-R	P_{out} (bar)	150.2	[58]
Int CL$_i$	T_{out} (°C)	40	[58]
Preheater	T_{out} (°C)	140	[58]
SPL-purge	ξ_{Purge} (%)	1.0	¤
PSA stream	y_{Ar} (%)	0.1	¤

*Adjusted to fit DWSIM. ¤Assumed value.

3.1.3 Complete model and load changes

By connecting each subsystem, the full plant-scale model was developed. A few global parameter values were assumed: Compressors, pumps, and blowers were defined with an efficiency of 75%, 86%, and 85%, respectively. Compressors were assumed to be powered by electric motor drives with an efficiency of 95% [64]. Apart from within the ammonia reactor, heat exchangers were assumed to have a pressure drop of 1 bar on each side. Electric heaters and coolers were defined with an efficiency of 100% and 90%, respectively.

The steam extraction flow should match the HX-cogen heat load to ensure complete feedwater vaporization and prevent inefficient over-extraction during load changes. When varying the ammonia production, the stream split ratio must therefore follow the load. To maintain focus on the broader system, a simplified modeling approach was utilized for the cogeneration heat exchanger rather than a detailed heat transfer analysis. The minimum mass flow rate of steam extraction can be found by:

$$|\dot{Q}_s| \geq |\dot{Q}_w| \tag{3.14}$$

$$\dot{m}_s(h_{h,in} - h_{h,out}) \geq \dot{m}_w(h_{c,out} - h_{c,in}) \tag{3.15}$$

$$\dot{m}_s \geq \dot{m}_w \frac{(h_{c,out} - h_{c,in})}{(h_{h,in} - h_{h,out})} \tag{3.16}$$

Where the heat transferred between the fluids is determined by the specific enthalpies before and after the heat exchange. Feed water is entering the heat exchanger at 1.35 bar and 25 °C, while steam is entering at 44.4 bar and 300 ° C. By forcing the steam to become fully saturated, using the heat exchanger as a condenser [3] for that stream, the outlet specific enthalpy of steam can be determined. At 44.3 bar, the specific enthalpy of saturated steam is $h_{sat,w}(44.3\text{bar}) = 1117.44\text{kJ kg}^{-1}$, and the saturation temperature is 256.4 °C [65]. A design outlet temperature of 230 °C for evaporated feed water was assumed, leaving a pinch point of $\Delta T_{pinch} = 256.4 - 230 = 26.4°\text{C}$ for the thermal driving force. The outlet specific enthalpy of water can then be determined, making the mass flow rate of steam a function of the mass flow rate of feed water. Because this is the minimum theoretical flow rate, a 1% margin was added, and equation 3.16 can be rewritten as:

$$\dot{m}_s(\dot{m}_w) \geq 1.01 \cdot \dot{m}_w \frac{(h_{c,out} - h_{c,in})}{(h_{h,in} - h_{h,out})} \tag{3.17}$$

Because the nuclear reactor steam system uses steam tables, the calculation of the heat exchange must match the same fluid package. Therefore, the electrolysis feed water stream through the cogeneration heat exchanger was set to use steam tables, and the outlet properties of the steam were transferred to the HTSE-feed via specification blocks, as seen in figure 3.2.

In this work, heat exchangers were assumed to be of shell-&-tube types, including the intercoolers in the multi-stage compression system. All heat exchangers were sized using the LMTD-method at nominal load, where the intercoolers were assumed to be cooled by water with an inlet and outlet temperature of 25 and 32 °C, respectively. The properties of the inlet and outlet streams of the HX-Cogen are described above, while the heat exchanger HX-return-O2 was dimensioned by defining a pinch temperature of 10 °C in DWSIM. The HX-ETHB was dimensioned to provide the ammonia reactor with the design inlet temperature of 140 °C. The overall heat transfer coefficients for different applications were applied to the relevant heat exchanger as presented in table 3.4.

Table 3.4: Applied overall heat transfer coefficients in heat exchangers [66][67]

Component	**Hot fluid**	**Cold fluid**	**U** $[\mathrm{W\,m^{-2}\,K^{-1}}]$
HX-return-O2	Anode outlet	SOEC feed	100
HX-EHB	Dry hydrogen	Recycle ammonia	300
Intercooler	Make-up gas	Water	300
HX-Cogen	Steam	Water	2600

The pressure losses in the interbed coolers and the ammonia chiller unit during part-load were assumed to scale with the square of the molar flow rate as described by [58]. The part-load strategy of the ammonia reactor was adopted from [68], where the operating pressure is kept constant. The part-load strategy of the SOEC stacks is assumed to be the same as in [57], where the individual stacks are kept on hot standby during load changes (varying amount of operating cells), allowing a constant temperature neutral operating point.

3.1.4 Preliminary analysis on equipment capacity

A preliminary analysis is necessary for the sizing of the plant, determining the capacity limits, and consequently the operating costs. System dimensioning is guided by the operating ranges for the cost correlations defined in table D.2. The preliminary analysis revealed that at a capacity of 700 tdp of ammonia, the power requirements for individual compressors in the multi-stage system slightly exceeded the 3,000 kW upper limit of the cost correlations by Turton et al. Conversely, at 600 tdp, all units remained within these scaling constraints. A nominal ammonia production capacity of 500 tpd was therefore established as the baseline, with the system designed to accommodate a 40% surge capacity, reaching a peak output of 700 tpd. By simulating the nominal production rate of 500 tdp, the required operating capacities of the equipment were determined. The pressure ranges, capacity units, and material selection for the different components are presented in appendix D.

3.2 Techno-economic model

The TEA presented in this work comprises the assessment of the profitability of the nuclear cogeneration plant in Europe based on the optimal thermal allocation for power and ammonia production. This includes the investigation of the revenues and operating costs from production.

As described in section 2.4.1, the future cost of SMRs is difficult to determine. In this study, the median values for the operating cost and OPEX multiplier in table 2.1 were assumed to represent the cost of the two E-SMR units in the system. The operating cost of the HTSE and HB system depends on the fixed capital cost as described in section 2.4.2, and the estimation approach is presented below. A $CEPCI_{2025}$ value of 800 was used for this.

For a fluidized-bed reactor, Turton et al. refers to the cost of a vessel with the same volume multiplied by five. This approach was taken for the ammonia reactor in this work, assuming a vertical pressure vessel, using the sum of the individual bed volumes, and rounding up to the nearest 10. A reactor diameter of 2 m was chosen and used to calculate the pressure factor from equation 2.21. The interbed coolers were considered to be included in the reactor cost. The ammonia flash separation unit was also considered a pressure vessel and dimensioned using the sizing option in DWSIM, which follows the Souders-Brown approach for a vertical vessel [69][70]. Some of the equipment costs are not mentioned in Turton et al. and are therefore found from other literature. These include the cost of the ammonia effluent chiller unit, gas topping heaters, PSA units for nitrogen feed and hydrogen purification, and the SOEC stacks. The cost of the SOECs was taken from [71] for stacks of electrolyte-supported cells, and a stack manufacturing rate of 200 MW/y. This yields a cost of approximately 150 \$/kWe (2021 value), converted to 192 €/kWe. The capital cost of the ammonia liquefaction chiller was taken from [72] for an ammonia process temperature of -25 °C. The electrolysis gas topping heater cost of 18.97 (2021\$/kWe) was found from [73]. Hydrogen dehydration following electrolysis was assumed using PSA, and the equipment cost was taken as 350 $\$/\mathrm{kg\,h^{-1}}$ (2024 value) of hydrogen [74]. At nominal load, the dry hydrogen stream after purification was 1.046 ks. The cost of nitrogen from PSA was based on data from different vendors [75][76][77] for a nitrogen purity of 99%. At nominal load, the ammonia production requires a volumetric flow rate of 0.889 $\mathrm{m^3\,s^{-1}}$ of nitrogen and inert. This corresponds to a large-scale nitrogen generator, and the price was assumed to be 50,000 \$. Lastly, the blowers were considered centrifugal radial fans.

The equipment material was selected based on compatibility with the respective stream properties and compositions. For components handling material streams containing hydrogen and/or ammonia, the stainless steel option in Turton et al. was assumed [78, 79]. Carbon steel (CS) was assumed for the electrolysis feed water pump and on the shell side of heat exchangers in contact with cooling water. For the HX-return-O2 recovering heat from the high-temperature pure oxygen, nickel based alloy was assumed [80] to handle these conditions.

In estimating the operational cost, the entire cogeneration plant was assumed to have an overall capacity factor of 90%. The cost of utilities C_{UT}, for the chemical process modules in this work, includes the cost of high-purity process water as feed water for electrolysis. The price of process water was based on [81], escalated to a 2025 value of 0.044 €/t. The cost of raw material and waste treatment is considered negligible, as no leakage is assumed, and the cost of radioactive waste is already included in the estimates by INL. The estimation of the required number of operators is based on the same approach as in [21], where a single operator is assumed to work an average of 49 weeks per year with five 8-hour shifts per week. Further assuming a 24h/d plant operation, the required number of operators to provide the corresponding number of shifts is approximately 4.5 operators. The operator's hourly wage was based on [81], escalated to 25.39€/h, and the number of process equipment amounts to 25. The final labour cost is therefore given as:

$$N_{operator} = 4.5 \cdot [(6.29 + 0.23 \cdot 25)^{0.5}] = 16.48 \rightarrow 17$$

$$C_{OL} = 17 \cdot 25.39(€/h) \cdot 49(week/y) \cdot 5(shift/week \cdot 8(h/shift) = 796230 \quad [€/y]$$

The annual profits from operations can be expressed as the sum of the production revenues from each commodity using the respective selling price subtracted my the respective cost of production. The pure oxygen byproduct of the hydrogen production process can also be sold as a commodity [28]. However, this was not considered in this cost analysis. It is important to note that the operating cost of ammonia production in equation 2.23 is expressed in cost per unit time, and needs to be converted to cost per unit production. Furthermore, because the generated electrical power is used internally, the net power sold to the grid is less than the power generated by the turbines:

$$\pi = \mathrm{P}_{el}\dot{W}_{net} - OPEX_{el}\dot{W}_{turb} + (\mathrm{P}_{\mathrm{NH_3}} - OPEX_{\mathrm{NH_3}})\dot{m}_{\mathrm{NH_3}} \quad [€/y] \tag{3.18}$$

Where P_{el} [€/MWh] and $P_{\mathrm{NH_3}}$ [€/ton] are the respective wholesale prices, while $OPEX_{\mathrm{NH_3}}$ and $OPEX_{el}$ are the production costs. By letting revenues of electricity equal the revenues of ammonia, a marginal break-even electricity price (BEP) can be defined as:

$$BEP \cdot \dot{W}_{net} - OPEX_{el} \cdot \dot{W}_{turb} = (\mathrm{P}_{\mathrm{NH_3}} - OPEX_{\mathrm{NH_3}})\dot{m}_{\mathrm{NH_3}} \tag{3.19}$$

$$BEP = \frac{(\mathrm{P}_{\mathrm{NH_3}} - OPEX_{\mathrm{NH_3}})\dot{m}_{\mathrm{NH_3}} + OPEX_{el}\dot{W}_{turb}}{\dot{W}_{net}} \quad [€/\mathrm{MWh}] \tag{3.20}$$

This defines the tipping point for operational decision-making and defines the electricity price threshold where the marginal profit from ammonia synthesis equals the potential revenue from electricity sales. Generally, if $P_{el} < BEP$ for a specific ammonia production capacity, the system should generally divert energy for ammonia production for increased profitability, and vice versa.

4 Results and analysis

In this section, the results from both the individual subsystem model validation, the complete simulation model, and the techno-economic analysis are presented. Differences in models, simulation methodology, and the resulting system capacities, efficiencies, and economic metrics are discussed. The retrieved data were evaluated using Python and Excel when necessary to facilitate a rigorous comparative analysis and enhanced data visualization.

4.1 Validation of main system components

Validation of secondary steam loop

A comparison between the simulation results of the reactor secondary steam loop model, without steam extraction in DWSIM and the reference model is presented in the table below:

Table 4.1: Comparison between results of the reference CYCLOP models, and DWSIM outputs

Parameter	Ref [54]	**DWSIM**	**Rel Diff %**
SG inlet temperature (°C)	163	162.81	0.12
SG mass flow (kg/s)	239.64	239.90	0.11
SG thermal input (MWth)	-	541.26	-
HPT outlet temperature (°C)	-	168.08	-
IPT outlet temperature (°C)	93.5	93.99	0.52
MS liquid mass flow (kg/s)	11.64	12.53	7.65
Total power output (MWe)	180.6	186.96	3.52

The most notable difference between the available data is the total power output, where the discrepancy might be attributed to both the increased thermal input to the SG and the fact that the model does not account for the same loss mechanisms. Based on the comparison, the steam cycle model was considered to behave within sufficient margins for reliable thermodynamic simulations.

Validation of SOEC model

At an operating voltage of 1.285V, M. Schiedeck et al reported an average current density of -0.759 $\mathrm{A\,cm^{-2}}$. In the present model, a current density of -0.762 $\mathrm{A\,cm^{-2}}$ was obtained under the same conditions. This represents a minor relative difference of 0.39%, a discrepancy likely attributable to the employment of a Fickian diffusion model rather than the more complex dusty-gas model used in the reference study. The comparison between these values suggests that the model is in a sufficient level of agreement with the reference model, thereby confirming its suitability for evaluating the thermodynamic performance of the SOECs.

Validation of ammonia reactor model

In their optimization, Fahra et al. set a lower limit of 21% for the ammonia fraction of the reactor effluent. This was set as a degree of freedom along with the heat exchanger UA-values in this work. DWSIM only allows for separate heat transfer coefficient and heat exchange area inputs. Therefore, the outlet temperatures of the hot stream in the ICs were defined as the inlet temperatures for beds 1 and 2 in the reference paper. DWSIM then calculated the U-values based on an input area and adjusted the outlet temperatures to reach a solution. The UA-value of the iFEHE from the reference was used as an input to the U-value in DWSIM, and then the heat transfer efficiency was adjusted to match the first bed inlet temperature of 357.4 °C. Pressure drops in all components at full load were set identically to the reference. Using the input parameters listed in table 3.3, the ammonia fraction versus the inlet and outlet temperatures of the beds at full load is presented in figure 4.1.

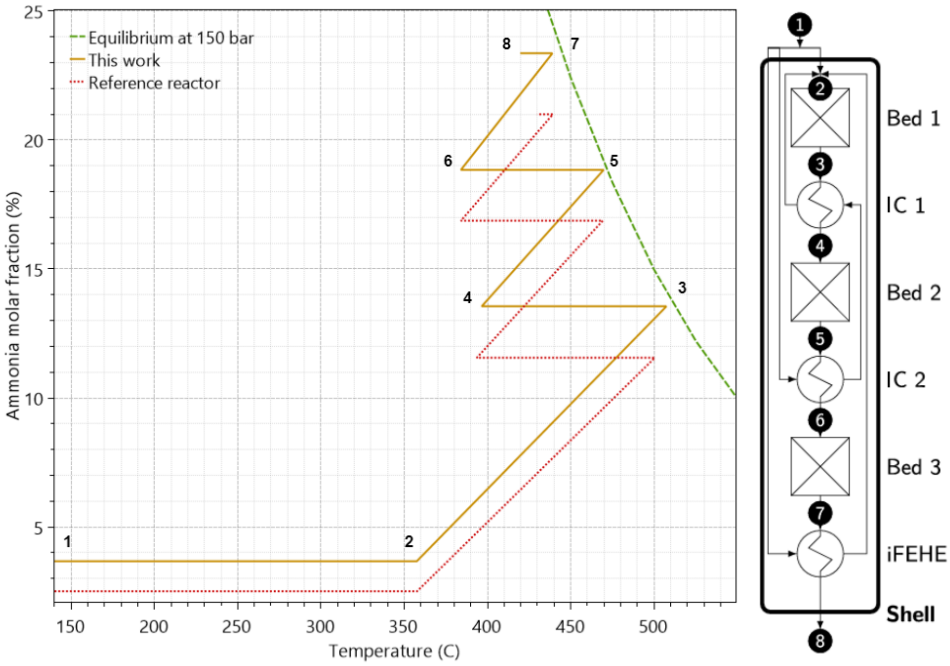


Figure 4.1: Ammonia fraction in stream versus inlet and outlet temperatures of each reactor bed at an operating pressure of 150 bar and a nominal production of 1356 tpd.

It was observed that the DWSIM reactor model behaves similarly to the reference model and is slightly more efficient. The reaction rate reaches closer to equilibrium in each bed before the gas is cooled, consequently achieving a higher outlet ammonia fraction. A comparison between the reactor model output is presented in table 4.2

Table 4.2: Comparison of output results ammonia between reactor models

Parameter	This work	Reference	Unit	Rel Diff %
T^{in}_{Bed1}	357.4	357.4	°C	0
T^{in}_{Bed2}	396.5	393.1	°C	0.86
T^{in}_{Bed3}	384.0	384.0	°C	0
UA_{IC1}	132.8	138.96	kW K^{-1}	0.44
UA_{IC2}	56.77	63.31	kW K^{-1}	10.33
UA_{iFEHE}	221.85	146.44	kW K^{-1}	51.50
$\text{y}^{out}_{\text{NH}_3}$	23.3	21.0	%	10.95

The model achieves results relatively close to the reference, except for the iFEHE heat transfer coefficient. However, this value differed substantially in the reference work from the original results as well, suggesting a high level of sensitivity to underlying assumptions or potential discrepancies in the reporting of the reference simulation. Fahra et al. mentions that a lower limit of 21% ammonia fraction is on the upper end of the range found in the literature, and attributes it to the purge-free system. The DWSIM model achieves an ammonia fraction of 23.3% of the reactor effluent, which might be attributed to unrealistically ideal heat management and conditions. The model was tested with an assumed argon fraction of 0.5% in the nitrogen stream from the ASU [82], corresponding to around 0.13% in the reactor feed with stoichiometric composition, and a purge stream of 1% [82] of the recycle. This resulted in an argon fraction between 1.2-1.4% in the reactor, and an ammonia fraction of 22.8% in the effluent. This configuration was further used in this work. Based on the comparison of models, the results demonstrate a sufficient level of agreement with the reference data, confirming that the current model is reliable tool for subsequent techno-economic analysis.

4.2 System dimensioning

In figure 4.2 the power production of E-SMR unit and the power consumption of the subsystems are plotted.

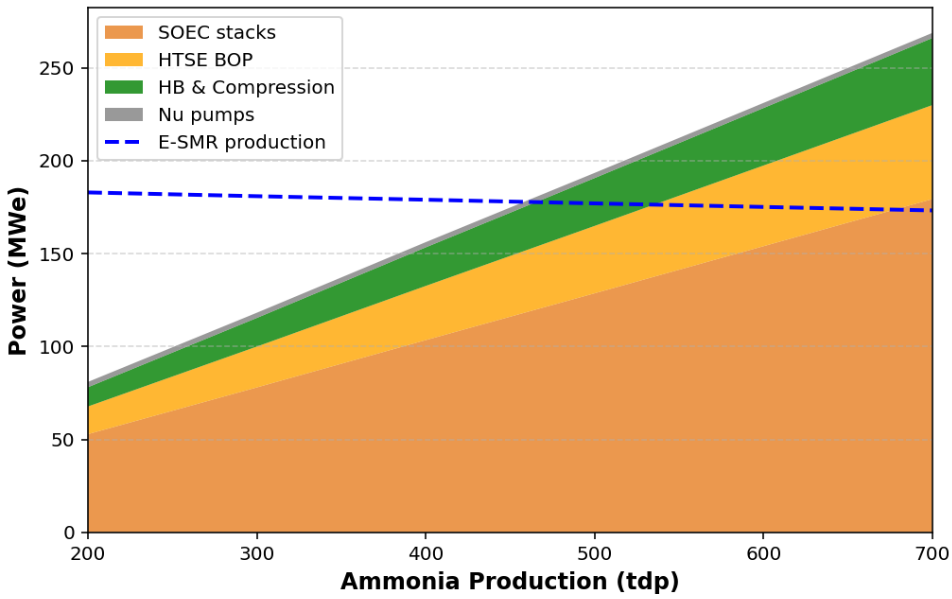


Figure 4.2: Electric power consumption of the SOEC stacks, balance of plant components in the HTSE system, HB process, and multi-stage compression system, as well as the power production by the thermally integrated SMR unit at different ammonia production loads.

It can be observed that the overall power consumption increases linearly with increasing load, where the SOEC stacks within the HTSE system were found account for approximately 78% of the total consumption at every production load. This is due to the maintained thermoneutral current density and the constant steam utilization in the SOECs, creating a linear relationship between required power input and increased hydrogen production. The power production from the simulated SMR unit decreases slightly from 182.8 to 173.2 MWe as steam is diverted for heat exchange. An intersection point between the power production and the total consumption of the system shows that one unit of the E-SMR alone can not supply enough power for the entire system. At an ammonia production rate of 453.8 t/d, the system's electrical demand reaches 177.9 MWe. This represents a critical operational threshold where total power generation is consumed internally, resulting in a net-zero balance with the external grid.

As the power consumption of the system requires at least two of the E-SMR units for the nuclear power system to be able to supply enough electric power at 600 and 700 tdp, a second E-SMR unit was integrated into the analysis to investigate the medium scale production. However, instead of thermodynamically modelling an additional steam loop, the power input from a second unit can be represented by mathematically adding the power output of $\dot{W}_{2.unit,turb}$ = 186.96 MWe with no cogeneration, and doubling the thermal input value from the steam generator. Hence, the additional pump work must be considered in the efficiency calculation (equation 4.1). Therefore, a second SMR unit is assumed to operate without thermal integration, supplying power to the grid and supplemental electricity to the ammonia synthesis process in instances where the primary unit output is insufficient. The global system efficiency is then defined as:

$$\eta_{cogen,2units} = \frac{\dot{W}_{net} + \mathrm{LVH}_{\mathrm{NH_3}} \cdot \dot{m}_{\mathrm{NH_3}}}{\dot{Q}_{Nuclear}} = \frac{\dot{W}_{1.unit,turb} + \dot{W}_{2.unit,turb} - \dot{W}_{internal} + \mathrm{LVH}_{\mathrm{NH_3}} \cdot \dot{m}_{\mathrm{NH_3}}}{\dot{Q}_{Nuclear}} \tag{4.1}$$

4.3 Cogeneration and efficiencies

By implementing two E-SMR units the nuclear reactors are able to supply sufficient amount of energy for scaling the ammonia production to 700 tdp. The gross power production decreases slightly as steam is used to evaporate water for steam electrolysis instead of expanding through the turbines.

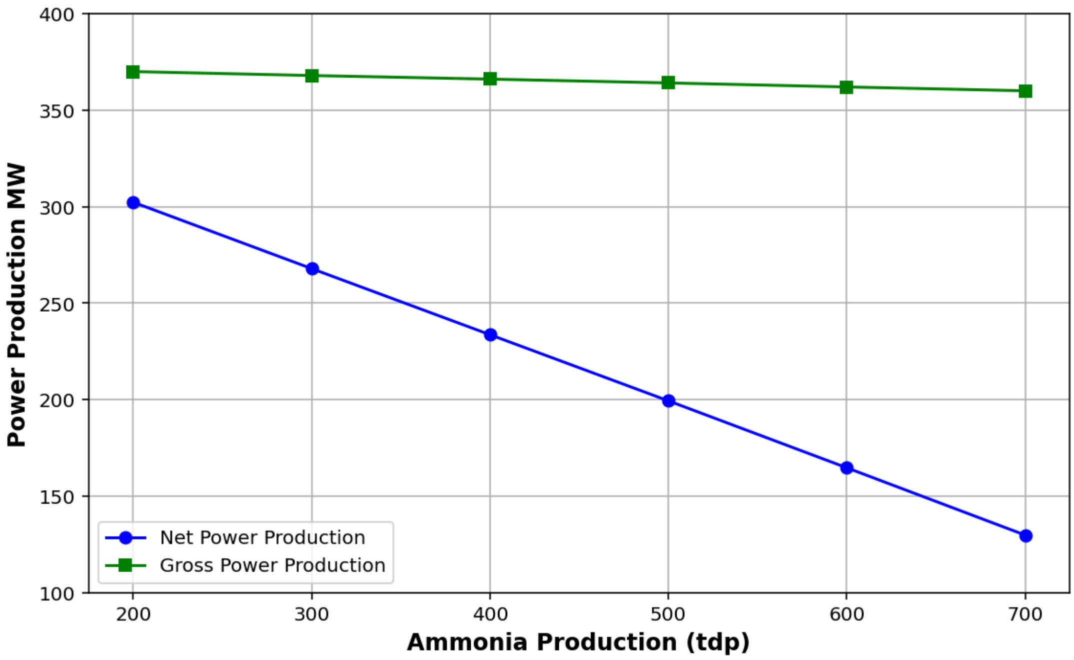


Figure 4.3: Power production as a function of ammonia production for two E-SMR units.

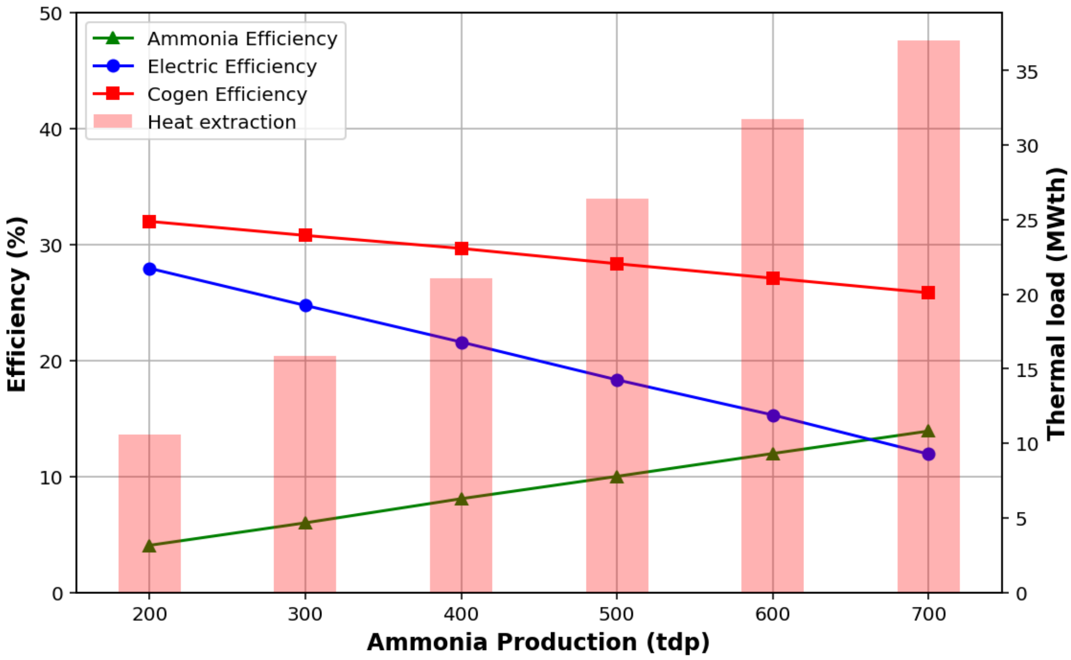


Figure 4.4: Thermal to electric, ammonia, and cogeneration efficiency at different production rates of ammonia and the corresponding heat extraction in the cogeneration heat exchanger.

In figure 4.4, the efficiencies of the system with two E-SMR units are presented for different ammonia production rates. As expected, the thermal to electric efficiency decreases with increasing production. However, the decrease is to such an extent that it is not offset by the increasing thermal to ammonia efficiency, leading to the decrease in cogeneration efficiency from 32% to 25.9%. [10] reported increasing cogeneration efficiency for increasing ammonia production loads using a VHTR. However, this type of reactor is able to provide much higher steam temperatures compared to a PWR-type SMR, and therefore might offer better heat utilization. Still, [11] found increasing heat utilization at higher hydrogen production loads using the PWR, VVER-1000 reactor. This suggests that the thermal utilization of the system in this study is insufficient. There are several factors that can potentially improve the cogeneration efficiency. Increasing the operating temperature and pressure of the steam electrolysis, or reducing the steam utilization can potentially reduce the overall energy consumption. For example by extracting the high pressure steam directly from the reactor secondary loop. Furthermore, there is potential for enhanced thermal integration to improve overall system efficiency. Specifically, the high-temperature oxygen byproduct generated by the SOEC stacks represents a high-grade heat recovery source. This was not, however, explored in this work.

4.4 Break-Even point and optimal production capacity

Moving on to the results of the techno-economic analysis. For a design load of 500 tdp of ammonia, and a capacity factor of 90%, the operating cost of the power and ammonia production respectively within the capacity window are presented in table 4.3

Table 4.3: System operating costs

Abbreviation		Value	Unit
Operating cost, 2 SMRs	(23,71 €/MWh)	69.23-67.37	M€/y
chemical plant operating cost	(843.9-241.2 €/t)	55.44-55.46	M€/y

The cost per ton of green ammonia are within reasonable margins according to [83]. A detailed breakdown of the underlying chemical plant costs is presented in Appendix D. These values were further used to investigate the profitability of the system in Europe.

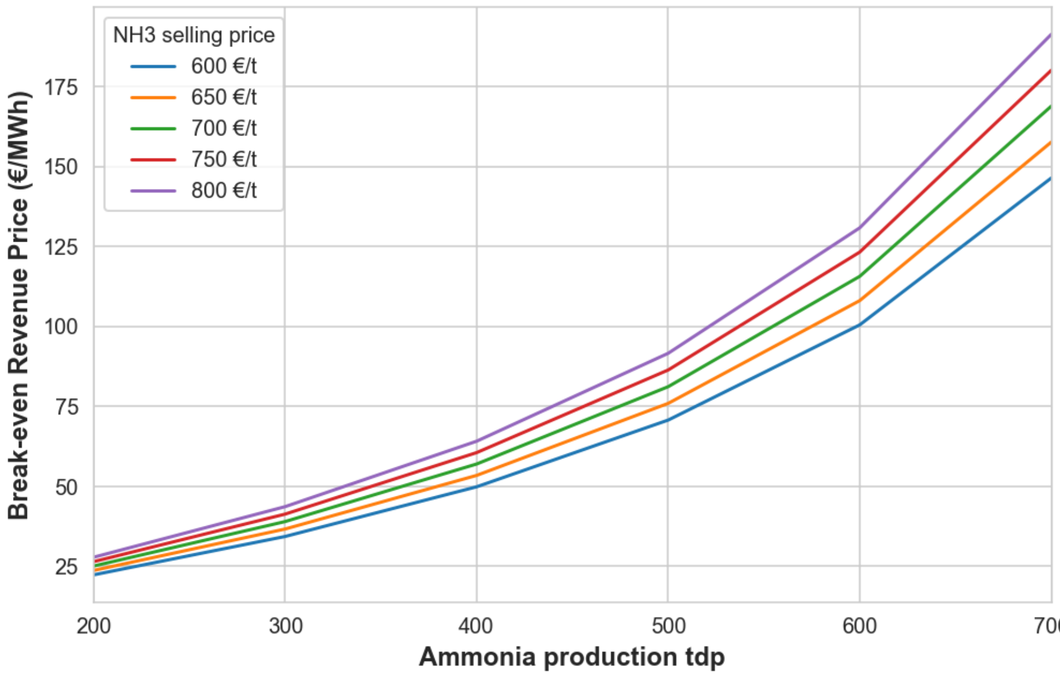
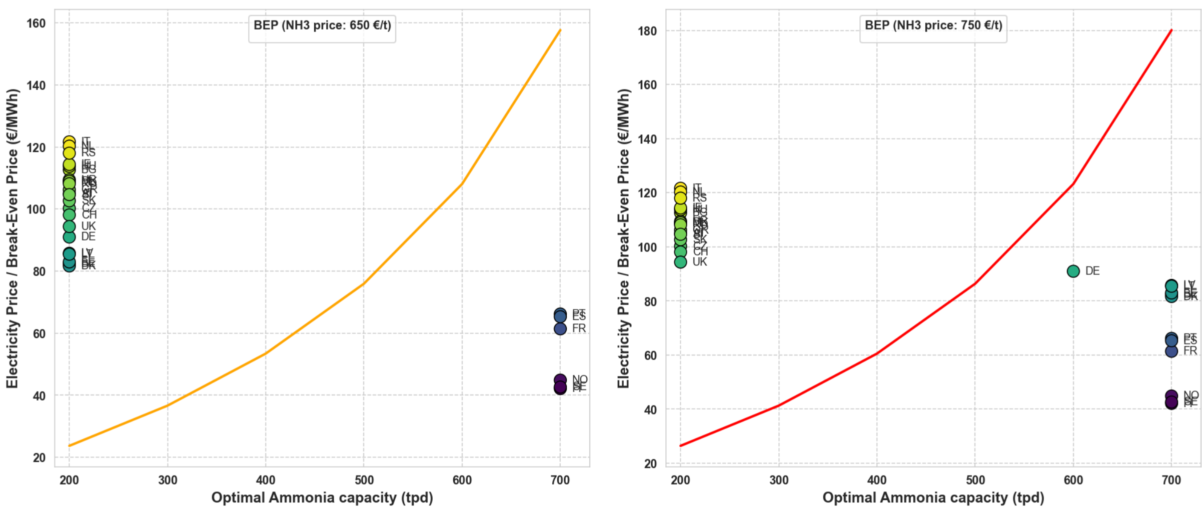


Figure 4.5: Break-even revenue electricity prices as a function of ammonia production load for five selected market prices of ammonia.

Figure 4.5 shows the break-even prices when revenues from electricity equal the revenues from ammonia, for five selected CFR ammonia prices in NW Europe, based on the market data presented in section 2.4.4. At lower ammonia production, differences in the market price have less significance to the break-even electricity price compared to at greater loads. Depending on whether the received electricity selling price is above or below the BEP curve at a specific ammonia capacity, the cogeneration plant can choose its mode of operation to maximize profits. However, because the ammonia synthesis is an energy intensive process, the opportunity cost is high because every MWh consumed to make ammonia is a MWh that could have been sold to the grid. Therefore, if an electricity prices is below the BEP at a certain capacity, it does not automatically dictate that ammonia production is profitable. The system must also account for the opportunity cost of the electricity consumed, and a lower ammonia capacity might be more profitable, which consequently lower the BEP. By using the listed median electricity prices in the European markets presented in table 2.4, the total revenues for each ammonia production load in each market can be calculated. It is then possible to determine the most profits the co-generation system can gain within the specific market, based on a given ammonia selling price, and its underlying marginal costs.



(a) (b)

Figure 4.6: Optimal ammonia production rate per country based on the respective median electricity price in 2025, and an ammonia selling prices of (a) 650 €/t, (b) 750 €/t. The lines shows the corresponding break-even prices for the system.

Figure 4.6, presents the optimal ammonia production capacities for the cogeneration system to maximize profits depending on the specific market and the ammonia spot price. The vertical delta between a country's position and the threshold line represents the comparative advantage between ammonia synthesis and electricity sales. Furthermore, the clustering of low-price markets at the 700 tpd mark (Finland to Portugal in figure 4.6a) indicates that in these regions, profitability is maximized through economies of scale, whereas in higher price markets, returns are maximized by minimizing production to capitalize on grid revenue. In higher price regions, such as in the Italy (IT) or the United Kingdom (UK), the opportunity cost of diverting energy to ammonia production is not strategic. For Nordic countries like Norway (NO), Sweden (SE), and Finland (FI), electricity is cheap, making the opportunity cost low. Therefore, the graph indicates that it is more profitable to convert cheap energy into a higher-value chemical, even at the lowest ammonia price levels. The optimization results generally demonstrate a binary behavior for most European markets because the system would obviously maximize the most profitable production option. However, as described in section 2.2, some capacity is required to avoid shutdown of operation. By observing when the selling price of ammonia increases, more markets favor increased ammonia production. There is also a narrow range where the opportunity cost of electricity reaches parity with the marginal profitability of increased ammonia synthesis, which shifts with increasing ammonia prices. In these scenarios, the Danish power market (DK), for example, represents a unique solution, where the power price creates a trade-off between chemical revenue and the opportunity cost of electricity, making 600 tdp the optimal production capacity at $P_{\mathrm{NH_3}} = 700$ €/t. This is further illustrated by looking at the annual profits in the figure below:

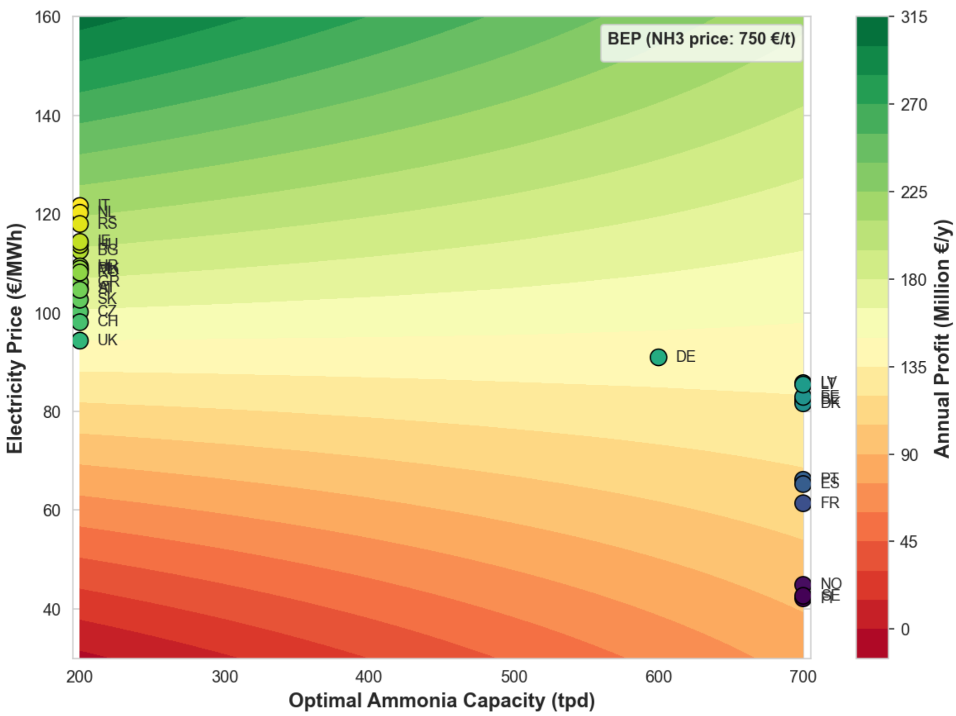
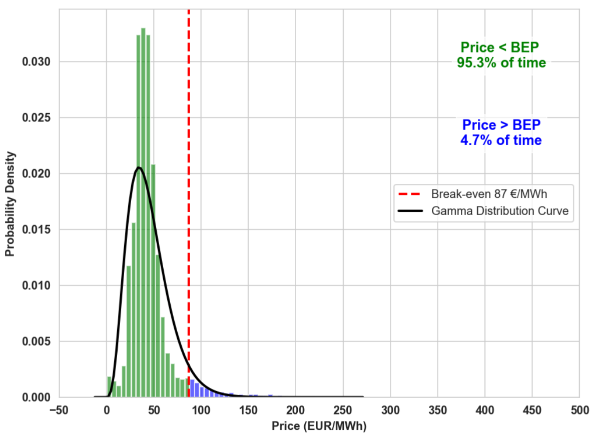


Figure 4.7: Contour plot of annual profits as a function of ammonia capacities, and median 2025 electricity prices per market for an ammonia price of 750 €/. Annual profits increase from red to green color grading.

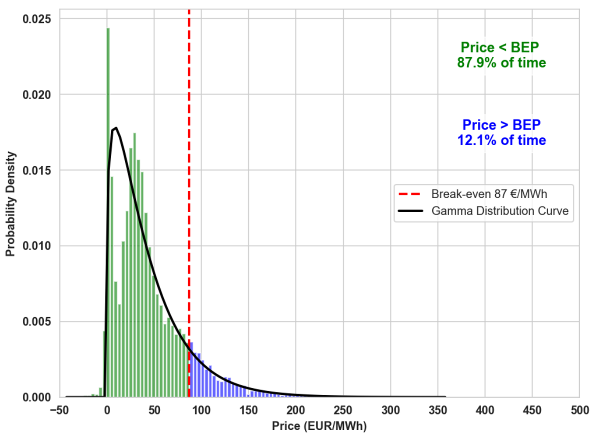
The green-most color grade from a horizontal line based on the respective power prices, dictates the optimal ammonia capacity for most annual returns. While these figures define the optimal ammonia capacity based on static average electricity prices, actual market conditions are characterized by significant temporal volatility. Consequently, evaluating the cumulative frequency for which each operational mode remains optimal provides a better understanding how the co-generation nuclear plant should operate.

4.5 Mode of operation based on market volatility

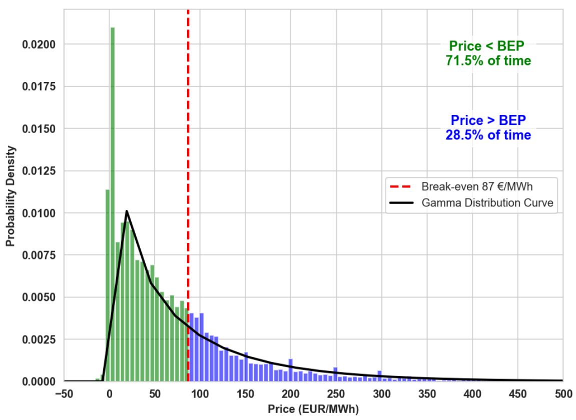
To investigate how often within a given time-frame the cogeneration system should produce the respective commodity to increase profits, it is possible to look at the frequency of the received power prices in different European markets. It is then necessary, to choose a BEP, reflecting the given selling price of ammonia. For this evaluation, a break-even revenue point of 87 €/MWh was selected, corresponding to a production capacity of 500 tpd at an ammonia selling price of 700 €/t, as illustrated by the green middle curve in figure 4.5. Historical day-ahead electricity prices in 2025 were retrieved from the ENTSO-E Transparency Platform and EMBER using Python, and aggregated into probability density histograms to evaluate the profitable hours of each mode of operation across selected markets.



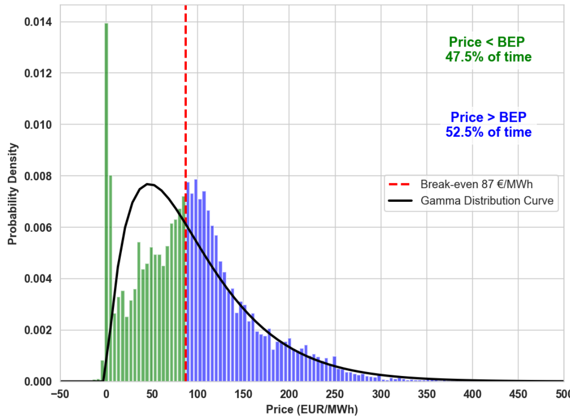
(a) Norway



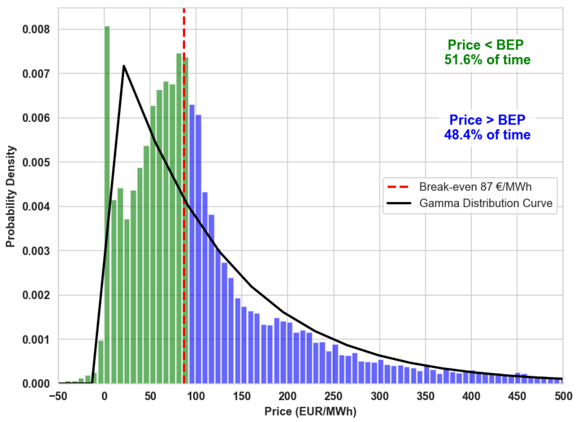
(b) Sweden



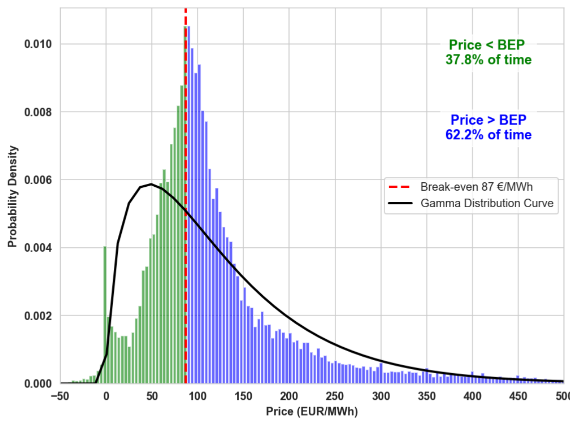
(a) Finland



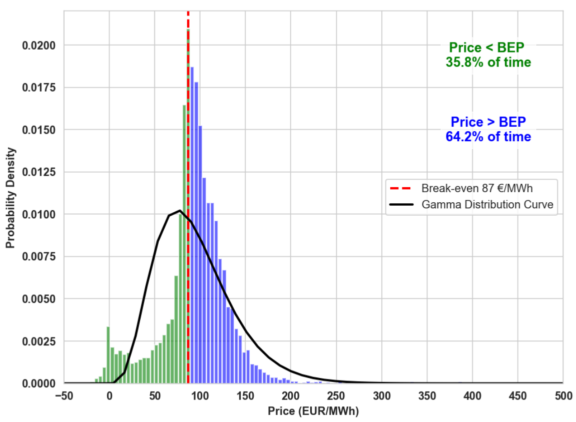
(b) Spain



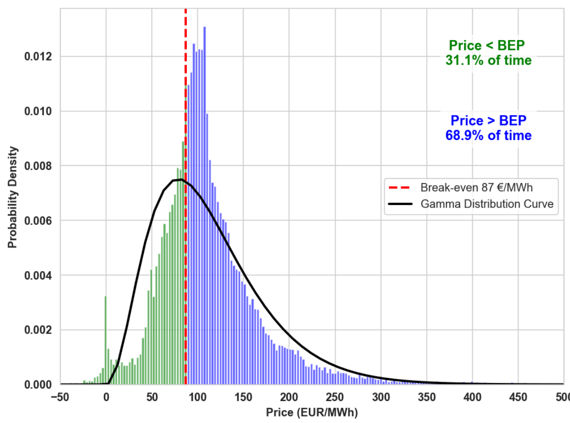
(c) France



(d) Germany/Luxembourg



(e) United Kingdom



(f) Poland

Figure 4.9: Probability density distribution of hourly wholesale electricity prices (€/MWh) from 01-2025 to 01-2026 across nine selected European markets retrieved from [84][46]. Prices for Norway and Sweden are aggregated across national bidding zones. The red dashed line denotes the break-even point of 87 €/MWh. Blue and green bars represent price frequencies below and above the BEP, respectively.

The distribution plots above, illustrates the percentage of power prices in each market that are profitable or not for ammonia production of 500 tdp and an ammonia price of 700 €/t. Or, equivalently, the percentage of hours over the year. However, it is important to note the mentioned transition zone, where the opportunity cost of the electricity becomes higher than the marginal profit of producing more ammonia, even if ammonia production is technically more profitable. Observing the distribution plots, the frequency of the day-ahead prices in the top row Nordic countries indicates that for this scenario, prioritizing ammonia production over 70% in Finland to over 90% in Norway of the time, would be more profitable. For the south and central European markets in the second row, the frequency of higher electricity prices indicates a slightly more favorable power mode of operation. Although the probability density reaches its maximum at 0 €/MWh in the Spanish and French markets, reflecting frequent periods of power exports, the cumulative area above the BEP is larger. This indicates that while the ammonia production could capture the most profitable individual hours, using the steam for power production should remain the dominant operational mode throughout the year. However, the share of hours per year with negative prices in Spain is expected to occur more frequently due to the increasing implementation of renewable energy technologies [85]. Electricity prices in the bottom row markets generally favor power over ammonia production, with the Polish market prices resulting in about 0.689×8760h/y ≈ 6036 power mode hours per year. An appreciation in the ammonia market price would shift the BEP toward higher electricity prices as seen in figure 4.5. This shift broadens the range of market conditions under which the ammonia production becomes more profitable, potentially favoring increased ammonia production capacities. Conversely, a decline in ammonia value would contract this range, rendering dedicated power generation more economically favorable. Scaling up production capacities would enhance the system's economic resilience to high electricity price volatility by allowing more power to be diverted to ammonia production rather than grid sales, thereby broadening the strategic operational range and making the system suitable in more European markets. Conversely, a smaller-scale system would be more restricted by lower ammonia capacity limits, making it more suitable in low electricity price markets.

4.5.1 Summary of results

Referring to the first research question 1 of this work, the results indicate that at small to medium-scale ammonia production (<500 tdp), one E-SMR unit is able to supply the entire cogeneration system with energy and electrical power, while at larger ammonia production capacities, the electrical demand of the system exceeds the single-unit generation threshold. However, for the ammonia production to be competitive within within most European markets other than in the Nordic, the results indicate that a production capacity of 600–700 tpd is required, necessitating the deployment of multiple SMR units. The system achieves low cogeneration efficiencies when increasing ammonia production, different from what is reported in the literature. However, this is also subject to the method of defining the system efficiency.

Regarding the second research question 2, the marginal break-even electricity price of the system ranges from 22.30 €/MWh at 200 tdp and an ammonia price of 600 €/t, to 191.35 €/MWh at 700 tdp and an ammonia price of 800 €/t. Furthermore, the results indicate that for a break-even revenue electricity price of 87 €/MWh, ammonia production is more profitable than power production over 90% of the operating hours in the Norwegian market, 80% in the Swedish, and 70% in the Finnish market. Analysis of the remaining selected European markets shows less profitable conditions for ammonia production at this BEP. However, this viability is sensitive to both power and ammonia market fluctuations, which are also interdependent markets.

5 Conclusion

In this work, a nuclear power and green ammonia cogeneration plant with a nominal capacity of 500 t_{NH_3} per day based on a 540 MWth European small modular reactor, high-temperature steam electrolysis using solid-oxide electrolysis cells, and the Haber-Bosch process was studied to analyze the system efficiencies and potential for increased profitability in different European electricity markets. The configuration and design of the cogeneration thermodynamic model was developed in the chemical process simulator DWSIM, based on validated systems of a nuclear reactor steam loop, steam electrolysis, and ammonia synthesis. Additionally, a techno-economic analysis was performed, including an estimation of the system's individual production costs of electricity and ammonia, and an investigation of the most profitable allocation of thermal energy based on retrieved day-ahead power prices in selected European markets. The analysis indicated that for the nuclear reactor system to supply sufficient energy for the entire plant at above approximately 453.8 ton per day, multiple SMR units are required. Cogeneration of power and ammonia offers the potential for an operational strategy that can provide increased profits, particularly in the Nordic European markets. At the nominal capacity, production of ammonia is generally more profitable at approximately 70% of the received market prices in the Norwegian, Swedish, and Finnish market, even at the cost of possibly lower system efficiencies. Other European markets

Ultimately, the findings suggest that nuclear-driven cogeneration of power and green ammonia using small modular pressurized water reactors has significant potential to increase annual profits while avoiding low or negative bids in volatile European markets.

Recommendation for future work

Based on the limitations and findings of this Techno-Economic and operational strategy analysis, the following areas are recommended for future research to refine the model's accuracy and expand its applicability.

- **Dynamic System simulation:** The current thermodynamic model is limited to steady-state analysis. Future research should implement a dynamic simulation to investigate system behavior during transient operations and changes in production loads.
- **Advanced Heat Integration and Parameter Optimization:** While this study utilizes fixed values for key thermodynamic parameters, these variables significantly influence efficiency and economics. Future work should incorporate a sensitivity analysis to identify and optimize these parameters for maximum performance.
- **Detailed SMR Cost Modeling:** This analysis employs generalized production costs for the Small Modular Reactor as an input. Future studies should develop a bottom-up cost estimation tailored specifically to the technological configuration of this cogeneration system.
- **Dynamic Market Sensitivity Analysis:** Future studies should expand upon the price assumptions by conducting a dynamic sensitivity analysis for operational expenditures, as well as power and ammonia selling prices. This should encompass a wide range of scenarios, including high-volatility and low-cost conditions, to better assess long-term project viability.
- **Impact of Green Policies and Incentives:** Future work should include an analysis of how potential environmental policies promoting fossil-free ammonia production affect revenue streams and overall project economics.

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A SOEC model parameters and script

Table A.1: SOEC model parameters [57]

Parameter	**Value**	**Unit**
B_{Ω}	$7.8247 \cdot 10^{11}$	$\mathrm{SK\,m^{-2}}$
$E_{act,\Omega}$	$8.0022 \cdot 10^{4}$	$\mathrm{J\,mol^{-1}}$
$\alpha_{\mathrm{H_2}}$	0.77251	-
$\gamma_{\mathrm{H_2}}$	$8.99769 \cdot 10^{6}$	$\mathrm{A\,K^{-1}\,m^{2}}$
a	0.0538348	-
b	0.48814	-
$E_{act,\mathrm{H_2}}$	$1.20052 \cdot 10^{5}$	$\mathrm{J\,mol}$
$\alpha_{\mathrm{O_2}}$	0.83035	-
$\gamma_{\mathrm{O_2}}$	$2.43410 \cdot 10^{8}$	$\mathrm{A\,K^{-1}\,m^{2}}$
m	0.18247	-
$E_{act,\mathrm{O_2}}$	$1.48142 \cdot 10^{5}$	$\mathrm{J\,mol}$
Thickness cathode/anode/electrode	$350/40/10$	µm
Cathode/anode porosity	$0.31/0.40$	-
Cathode/anode turtuosity	2.52	-
Cathode/anode mean pore radius	1.75	µm
Cell dimension	0.04	$\mathrm{m^2}$
Steam Utilizaton	70	%
ΔP_{stack}	0.04	bar

```
import math

# Define input and output material streams
Feed = ims1
vapor_out = oms1
O2_out = oms4
O2_in = ims5
Energy_in = ies1
Heat_flow = oes1

# variables for input stream
T = Feed.GetTemperature()
P = Feed.GetPressure() #cathode pressure
P_anode = O2_in.GetPressure() #Anode pressure
moleflow_Feed = Feed.GetMolarFlow()
Feed_massflow = Feed.GetMassFlow()
In_moleflow_O2 = O2_in.GetMolarFlow()
Feed_density = Feed.GetPhase("Vapor").Properties.density # kg/m3
comp_feed = Feed.GetOverallComposition()
H2O_frac = comp_feed[0] #The position of the compound in list of the material stream
H2_frac = comp_feed[2]

# Partial pressures and molar flows
p_H2 = P*H2_frac
p_H2O = P*H2O_frac
p_O2 = P
in_molflow_H2O = moleflow_Feed*H2O_frac
in_molflow_H2 = moleflow_Feed*H2_frac

# General constants
P0 = 101325 # [Pa]
H_std = -241.8264 # kJ/mol
M_steam = 18.01528/1000 # kg/mol

# Electrochemical constants
B_ohm = 7.8247e11 # S K/m2
E_act_ohm = 8.0022e4 # J/mol
alpha_H2 = 0.77251
gamma_H2 = 8.99769e6 # A*m2/K
a = -0.0538348
b = 0.48814
E_act_H2 = 1.20051e5 # J/mol
alpha_O2 = 0.83035
gamma_O2 = 2.43410e8 # A m2/K
m = 0.18247
E_act_O2 = 1.48142e5 # J/mol

F = 96485
R = 8.314 # J/mol-k
z = 2

# Electrolysis cell constants
t_c = 350e-6 # cathode thickness m
t_a = 40e-6 # anode thickness m
t_e = 10e-6 # um
pore_c = 0.31 # Cathode porsity
pore_a = 0.40 # anode porsity
tortuosity = 2.52
r_pore = 1.75e-6
P_drop = 0.04e5 # Pressure drop over stack


#Shomate equations to define properties of cell reaction

def Cp_schomate(T, a, b, c, d, e, f, g, h):
    t = T/1000
    Cp = a + b*t + c*math.pow(t,2) + d*math.pow(t,3) - e/math.pow(t,2) # J/mol*K
    return Cp

def H_shomate(T, a, b, c, d, e, f, g, h):
    t = T / 1000
    H = a*t + b*math.pow(t,2)/2 + c*math.pow(t,3)/3 + d*math.pow(t,4)/4 - e/t + f - h
```

Figure A.1: SOEC script page 1

```
        # kJ/mol
    return H

def S_shomate(T, a, b, c, d, e, f, g, h):
    t = T / 1000
    S = a*math.log(t) + b*t + c*math.pow(t,2)/2 + d*math.pow(t,3)/3 - e/(2*math.pow(t,2)) + g  # J/mol*K
    return S

# ------------------------------------------------------------
# Temperature ranges and coefficients for schomate equation
# ------------------------------------------------------------
coeffs = {
    "H2O": {
        (500, 1700): [30.09200, 6.832514, 6.793435, -2.534480, 0.082139, -250.8810, 223.3967, -241.8264],
    },
    "H2": {
        (298, 1000): [33.066178, -11.363417, 11.432816, -2.772874, -0.158558, -9.980797, 172.707974, 0.0],
        (1000, 2500): [18.563083, 12.257357, -2.859786, 0.268238, 1.977990, -1.147438, 156.288133, 0.0],
    },
    "O2": {
        (100, 700): [31.32234, -20.23531, 57.86644, -36.50624, -0.007374, -8.903471, 246.7945, 0.0],
        (700, 2000): [30.03235, 8.772972, -3.988133, 0.788313, -0.741599, -11.32468, 236.1663, 0.0],
    }
}


def get_HSCp(T, species):
    for (Tmin, Tmax), params in coeffs[species].items():
        if Tmin <= T <= Tmax:
            return H_shomate(T, *params), S_shomate(T, *params), Cp_schomate(T, *params)
    (Tmin, Tmax), params = list(coeffs[species].items())[-1]
    return H_shomate(T, *params), S_shomate(T, *params), Cp_schomate(T, *params)

H_H2O, S_H2O, Cp_H2O = get_HSCp(T, "H2O")
H_H2,  S_H2, Cp_H2  = get_HSCp(T, "H2")
H_O2,  S_O2, Cp_O2  = get_HSCp(T, "O2")

# ------------------------------------------------------------
# Compute reaction properties
# H2O(g) → H2(g) + ½O2(g)
# ------------------------------------------------------------

H_r = (H_H2 + (1/2)*H_O2) - H_H2O - H_std #[KJ/mol]
S_r = (S_H2 + (1/2)*S_O2) - S_H2O  #[J/mol-K]
Gibbs_r = H_r - S_r*T/1000 #[kJ/mol]

V_th = H_r*1000/(F*z)

# --- 1. Voltage Calculations ---
# Ohmic

R_ohm = (T/B_ohm)*math.exp(E_act_ohm/(R*T))
V_ohmic = -j * R_ohm

# Defining effectiv diffusion coefficient from source: https://www.mdpi.com/2227-7390/11/2/366
def get_Deff(T, P, species):

    # P in Pa, T in K

    if species == "H2O":

        porosity = 0.31
        Mi, Mj = 18.015, 2.016   # g/mol for Fuller
        Vi, Vj = 6.12, 13.1         # Fuller volumes
```

Figure A.2: SOEC script page 2

```
    elif species == "H2":

        porosity = 0.31
        Mi, Mj = 2.016, 18.015  # g/mol for Fuller
        Vi, Vj = 6.12, 13.1         # Fuller volumes

    else:

        porosity = 0.40
        Mi, Mj = 31.998, 31.998 # g/mol for Fuller
        Vi, Vj = 16.3, 16.3


    # 3. Fuller Equation

    M_term = math.sqrt(1/Mi + 1/Mj)

    V_term = math.pow(math.pow(Vi, 1/3) + math.pow(Vj, 1/3), 2)

    D_bin = (3.1976e-4 * math.pow(T, 1.75) * M_term) / (P * V_term)


    # Knudsen Diffusion (Function of pore radius) Mi in kg/mol


    D_kn = (2/3) * r_pore * math.sqrt((8 * R * T) / (math.pi * Mi/1000))



    # Effective Diffusion (Adjusted for electrode Porosity/Tortuosity)

    D_eff = (porosity / tortuosity) * (1.0 / (1.0/D_bin + 1.0/D_kn))

    return D_eff



# Constants for diffusion
D_H2O_eff = get_Deff(T, P, "H2O")
D_H2_eff = get_Deff(T, P, "H2")
D_O2_eff = get_Deff(T, P, "O2")

# Cathode TPB Pressures (H2 is produced, H2O is consumed)
p_H2_TPB = p_H2 + (R * T * t_c / (z * F * D_H2_eff)) * abs(j)
p_H2O_TPB = p_H2O - (R * T * t_c / (z * F * D_H2O_eff)) * abs(j)

# Anode TPB Pressure (O2 is produced)
p_O2_TPB = p_O2 + (R * T * t_a / (4 * F * D_O2_eff)) * abs(j)


# Exchange Current Densities
j0_H2 = gamma_H2*T*math.pow(p_H2_TPB/P0, a)*math.pow(p_H2O_TPB/P0, b)*math.exp(-E_act_H2/(R*T))
j0_O2 = gamma_O2*T*math.pow((p_O2_TPB/P0),m)*math.exp(-E_act_O2/(R*T))

# Nernst potential
V_rev = (1000*Gibbs_r)/(z*F) + (R*T/(z*F))*math.log((p_H2_TPB/p_H2O_TPB)*math.pow(p_O2_TPB/P0,0.5))

# --- 3. Newton-Raphson Solver for Activation Overpotential ---
def solve_eta(j_target, j0, alpha, T):
    # Initial guess for eta
    eta = -0.01
    coeff = (z * F) / (R * T)

    for i in range(100):
        # Butler-Volmer: f(eta) = j0 * (exp(a*c*eta) - exp(-(1-a)*c*eta)) - j_target
        term1 = math.exp(alpha * coeff * eta)
        term2 = math.exp(-(1 - alpha) * coeff * eta)
        f = j0 * (term1 - term2) - j_target
```

Figure A.3: SOEC script page 3

```
        # Derivative: f'(eta) = j0 * coeff * (alpha*term1 + (1-alpha)*term2)
        df = j0 * coeff * (alpha * term1 + (1 - alpha) * term2)

        eta_new = eta - f / df
        if abs(eta_new - eta) < 1e-6:
            return eta_new
        eta = eta_new
    return eta

eta_act_H2 = solve_eta(j, j0_H2, alpha_H2, T)
eta_act_O2 = solve_eta(j, j0_O2, alpha_O2, T)

V_op = V_rev - j*R_ohm - eta_act_H2 - eta_act_O2



# molar balance
I_cell = abs(j)*A # current per cell / A in cm2
Target_H2_prod = X_su*in_molflow_H2O
I_tot = (F*z)*Target_H2_prod
N_cells = I_tot/I_cell     # Total number of cells required for the production
O2_prod = Target_H2_prod/2
Power = I_tot*V_op/1000 # kW


Q = I_cell*A*N_cells*(V_op - V_th)/1000 # Waste heat in KW that needs to be added/removed
Cp_H2O_unit = Cp_H2O/(M_steam*1000) # Specific heat capacity in KJ/Kg*K

# Set output material streams H2
total_out_H2 = Target_H2_prod + in_molflow_H2
total_out_H2O = (1-X_su)*in_molflow_H2O
totflow_O2 = O2_prod + In_moleflow_O2

# Set output hydrogen and steam stream
vapor_out.SetPressure(P - P_drop)
vapor_out.SetTemperature(T)
vapor_out.SetOverallCompoundMolarFlow("Hydrogen", total_out_H2)
vapor_out.SetOverallCompoundMolarFlow("Water", total_out_H2O)

# Set output material streams O2
M_enthalpy_O2 = H_O2/0.031998
O2_out.SetPressure(P_anode - P_drop)
O2_out.SetMassEnthalpy(M_enthalpy_O2)
O2_out.SetOverallCompoundMolarFlow("Oxygen", totflow_O2)

# Set energy streams
Energy_in.SetEnergyFlow(Power)
Heat_flow.SetEnergyFlow(Q)
```

Figure A.4: SOEC script page 4

B Multi-stage compression

If all the compressors in a multi-stage compression system have the same isentropic efficiency, a simple formula exists for an optimal equal compression ratio for each stage. For a compression ratio between intermediate pressures $r_{i,i+1} = \frac{P_{i+1}}{P_i}$, it can be observed that the product of all ratios is equal to the total pressure ratio [86]:

$$\prod_{i=1}^{n} r_{i,i+1} = \frac{P_2}{P_1}\frac{P_3}{P_2}...\frac{P_{n+1}}{P_n} = \frac{P_{n+1}}{P_1} = r_t \tag{B.1}$$

If an ideal gas is assumed, and intercooling between consecutive compression stages brings the gas back to the initial temperature T_1, it can be shown that the optimized work is:

$$\prod_{i=1}^{n} r = r^n = r_t \Rightarrow \underline{r = r_t^{(1/n)}} \tag{B.2}$$

However, to account for equal pressure drops ΔP in the intercoolers after each compression stage, the equation must be modified. The first compression stage $R_{1,2}$ is not really affected by an intercooler pressure drop, but the subsequent stages are:

$$\begin{aligned}\prod_{i=1}^{n} R_{i,i+1} = \frac{P_2}{P_1}\frac{P_3}{(P_2-\Delta P)}\frac{P_4}{(P_3-\Delta P)}...\frac{P_{n+1}}{(P_n-\Delta P)} &= \frac{P_{n+1}}{P_1}\prod_{i=2}^{n}\frac{P_i}{(P_i-\Delta P)} \\ &= \frac{P_{n+1}}{P_1}\prod_{i=2}^{n}\left(\frac{1}{1-\frac{\Delta P}{P_i}}\right)\end{aligned} \tag{B.3}$$

This adds a pressure correction factor that becomes unity when $\Delta P = 0$. Letting $\varepsilon_i = \left(1 - \frac{\Delta P}{P_i}\right)$, the compression ratios can be expressed as:

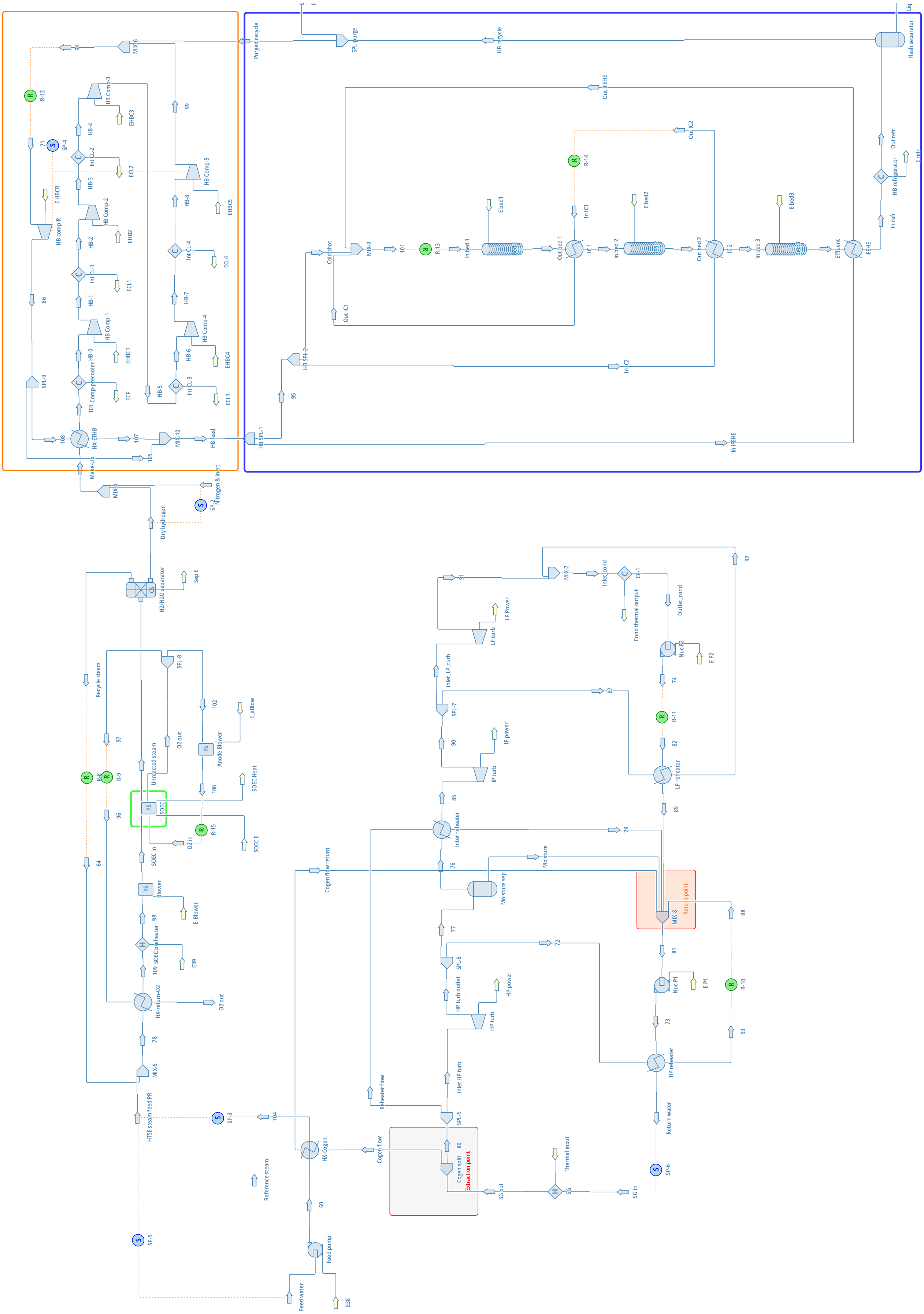
$$\prod_{i=1}^{n} R_{i,i+1} = \frac{P_{n+1}}{P_1 \prod_{i=2}^{n} \varepsilon_i} \tag{B.4}$$

Finally, the equation for the equal pressure ratio with intercooler pressure drop becomes:

$$\boxed{R = \left(\frac{P_{n+1}}{P_1 \prod_{i=2}^{n} \varepsilon_i} \right)^{(1/n)}} \tag{B.5}$$

To solve this, one needs to find the pressure stages P_i in the correction factor through $P_i = R \cdot P_{i-1}$, which gives a recurring relation and is easily solved numerically.

C Flowsheet of complete nuclear cogeneration system



D Equipment cost factors and constants

Table D.1: B1 and B2 equipment-specific constants

Equipment Type	Description	B1	B2
Heat Exchanger	Floating head	1.63	1.66
Process vessel	Horizontal	1.49	1.52
	Vertical	2.25	1.82
Pumps	Centrifugal	1.89	1.35

Table D.2: Equipment design capacities and material of construction

Equipment	Description	Type	Design capacity	MOC	F_M
Compressor	HB comp-1	Centrifugal	2169.37 kW	SS	5.8
	HB comp-2	Centrifugal	2172.53 kW	SS	5.8
	HB comp-3	Centrifugal	2179.84 kW	SS	5.8
	HB comp-4	Centrifugal	2197.42 kW	SS	5.8
	HB comp-5	Centrifugal	2242.14 kW	SS	5.8
	HB comp-R	Centrifugal	47.86 kW	SS	5.8
Drive	Drive C-1	Enclosed	2283.55 kW	-	1.5
	Drive C-2	Enclosed	2286.87 kW	-	1.5
	Drive C-3	Enclosed	2294.57 kW	-	1.5
	Drive C-4	Enclosed	2313.07 kW	-	1.5
	Drive C-5	Enclosed	2360.15 kW	-	1.5
	Drive C-R	Enclosed	50.38 kW	-	1.5
Heat exchanger	HX-Cogen	Floating head	74 m^2	CS-shell/SS-tube	1.81
	HX-ETHB	Floating head	56 m^2	SS-shell/SS-tube	2.73
	HX-return-O_2	Floating head	400 m^2	Ni-shell/Ni-tube	3.73
	Int CL-1	Floating head	163 m^2	CS-shell/SS-tube	1.81
	Int CL-2	Floating head	164 m^2	CS-shell/SS-tube	1.81
	Int CL-3	Floating head	165 m^2	CS-shell/SS-tube	1.81
	Int CL-4	Floating head	167 m^2	CS-shell/SS-tube	1.81
	Comp-precooler	Floating head	222 m^2	CS-shell/SS-tube	1.81
Vessel	Ammonia reactor	Vertical	90 m^3	SS clad	1.74
	Flash separator	Vertical	6 m^3	SS clad	1.74
Tray	In flash-separator	Demister pad	2.04 m^2	SS	1
Pumps	FW pump	Centrifugal	0.37 kW	CS	1
Fan	Cathode blower	Centrifugal radial fan	160 kW	SS	5.8
	Anode blower	Centrifugal radial fan	11.43 kW	CS	5.8
Chiller	Ammonia liquefyer	-	31824.18 kW	-	-
PSA	Nitrogen PSA	-	352 $kg\,h^{-1}$	-	-
	Hydrogen dryer	-	0.889 $m^3\,s^{-1}$	-	-
Heater	SOEC preheat	Electric	7558.78 kW	-	-
SOEC stacks	-	Planar	128674.67 kW	-	-

E Equipment design pressures

Table E.1: Design pressures and corresponding pressure factor constants.

Equipment	**Description**	**C1**	**C2**	**C3**	**Pressure Range (barg)**
Compressor	HB comp-1	-	-	-	-
	HB comp-2	-	-	-	-
	HB comp-3	-	-	-	-
	HB comp-4	-	-	-	-
	HB comp-5	-	-	-	-
	HB comp-R	-	-	-	-
Drive	Drive C-1	-	-	-	-
	Drive C-2	-	-	-	-
	Drive C-3	-	-	-	-
	Drive C-4	-	-	-	-
	Drive C-5	-	-	-	-
	Drive C-R	-	-	-	-
Heat exchanger	HX-Cogen	-0.00164	-0,00627	0.0123	149 (tube only)
	HX-ETHB	-0.00164	-0,00627	0.0123	149 (tube only)
	HX-return	0.03881	-0.11272	0.08183	0.12
	Int CL-1	-0.00164	-0,00627	0.0123	149 (tube only)
	Int CL-2	-0.00164	-0,00627	0.0123	149 (tube only)
	Int CL-3	-0.00164	-0,00627	0.0123	149 (tube only)
	Int CL-4	-0.00164	-0,00627	0.0123	149 (tube only)
	Comp-precooler	-0.00164	-0,00627	0.0123	149 (tube only)
Vessel	Ammonia reactor	-	-	-	149 (Fp_{vessel})
	Flash separator	-	-	-	149 (Fp_{vessel})
Tray	In flash-separator	-	-	-	149
Pumps	FW pump	-	-	-	-
Fan	Cathode blower	0.00	0.209	-0.033	0.02
	Anode blower	0.00	0.209	-0.033	0.04
Chiller	Ammonia liquefyer	-	-	-	149
PSA	Nitrogen PSA	-	-	-	4
	Hydrogen dryer	-	-	-	0
Heater	SOEC preheat	-	-	-	0.12
SOEC stacks	-	-	-	-	0.05

F Cost of equipment

Table F.1: Bare Module Cost Breakdown of Equipment

Equipment	Description	Base Bare Module Cost (€)	Bare Module Cost (€)
Compressor	HB Comp-1	2 991 920,83	6 278 665,97
	HB Comp-2	2 994 862,12	6 284 838,39
	HB Comp-3	3 001 658,70	6 299 101,28
	HB Comp-4	3 017 961,38	6 333 313,11
	HB Comp-5	3 059 164,48	6 419 779,48
	HB Comp-R	119 111,00	249 959,21
Drive	Drive for Comp 1	466 425,66	466 425,66
	Drive for Comp 2	466 548,69	466 548,69
	Drive for Comp 3	466 831,98	466 831,98
	Drive for Comp 4	467 505,86	467 505,86
	Drive for Comp 5	469 173,94	469 173,94
	Drive for Comp R	55 253,52	55 253,52
Heat exchanger	HX-Cogen	163 535,37	808 840,87
	HX-ETHB	46 836,34	310 592,09
	HX-return O2	134 230,72	1 572 917,97
	Int CL-1	72 782,05	337 309,02
	Int CL-2	72 904,12	341 255,32
	Int CL-3	73 163,17	347 067,02
	Int CL-4	73 679,34	356 039,69
	Comp precooler	87 573,23	404 409,11
Vessel	Ammonia reactor (incl. 3 beds)	686 329,59	70 722 095,72
	Amm separator	18 405,65	1 440 081,75
Tray	Demister	5 913,44	24 067,71
Refrig.	HB refrig	3 056 662,56	3 056 662,56
Blower	Anode Blower	532 897,82	492 387,25
	Blower (Cathode)	28 690,85	21 375,47
Pump	Feed pump (inc drive)	5 600,79	5 600,79
Heater	SOEC feed topping heater	183 539,27	183 539,27
PSA	H2 purifier (PSA)	1 476 034,56	1 476 034,56
	ASU unit (PSA)	56 000,00	56 000,00
Electrolysis	SOECs (Stacks)	24 705 336,64	24 705 336,64

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